

Artificial intelligence digital employees and sustainable innovation in online retail: The mediating role of ambidextrous green innovation and the moderating role of ethical anxiety

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ABSTRACT

The proliferation of artificial intelligence (AI) is rapidly transforming businesses across industries, driving an increased focus on sustainable innovation for long-term competitiveness. Within the online retail sector, AI-powered digital employees are emerging as key players in this transformation. This study investigates how AI digital employee adoption influences future sustainable innovation performance in online retail, examining the mediating roles of ambidextrous green innovation and the moderating effect of ethical anxiety. Drawing on the Antecedents-Behaviour-Consequences (ABC) model, we propose an Adoption-Innovations-Future Performance (AIF) framework to examine these relationships in the context of TikTok online retailers. Using a two-wave time-lagged survey of 302 online retailers, we find that digital employee adoption positively affects exploitative and exploratory green innovation, which in turn enhances future sustainable innovation performance. Digital employee ethical anxiety negatively moderates the relationship between adoption and green innovation but strengthens the positive link between adoption and future performance. The fsQCA results reveal three configurations leading to high sustainable innovation performance. This research contributes to the literature on digital transformation in retail and consumer services by elucidating the complex mechanisms through which digital employees influence organizational outcomes. The findings offer important implications for retailers seeking to leverage digital technologies for sustainability in their operations and customer service.

1. Introduction

The rapid proliferation of digital technologies is fundamentally transforming business operations and reshaping organizational capabilities across industries, with particularly profound implications for the retail and consumer services sectors (Bilal et al., 2024; Brüns and Meißner, 2024; Sun et al., 2024; Vhatkar et al., 2024; Wang and Zhang, 2024). According to a recent McKinsey Global Survey (2024), 65% of organizations are now regularly using generative AI, reflecting high expectations for the technology's impact on productivity and innovation within companies. The World Economic Forum (2023) estimates that by 2027, AI and automation could displace up to 25% of current jobs, necessitating a significant shift in workforce skills and organizational strategies. Furthermore, the International Monetary Fund (IMF) reports that globally, nearly 40% of jobs are at risk due to AI, with this figure rising to 60% in advanced economies (Cazzaniga et al., 2024). PwC's

2024 Global CEO Survey indicates that 69% of CEOs anticipate AI will necessitate new skills from their workforce, with this figure rising to 87% among those who have already deployed AI (PwC, 2024). Gartner predicts that by 2028, at least 15% of day-to-day work decisions will be made autonomously through agentic AI, up from 0% in 2024, dramatically increasing the adaptability of software systems to complete a wide variety of tasks (Gartner, 2024). At the forefront of this digital revolution is the emergence of digital employees, which are AI-powered virtual agents that can perform cognitive tasks traditionally reserved for human workers in customer-facing roles (Le et al., 2024). As retailers and service providers increasingly integrate these digital employees into their workflows, it raises critical questions about the implications for innovation processes, customer experiences, and sustainable performance outcomes (Singh et al., 2024).

Sustainable innovation refers to the development of new products, services, processes, or business models that not only deliver economic

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value but also contribute positively to environmental and social outcomes (Ardito, 2023; Arshad et al., 2024; Rauter et al., 2023). In the context of online retailing, sustainable innovation goes beyond traditional innovation by explicitly integrating environmental stewardship and social responsibility into innovation activities. This includes initiatives such as reducing carbon footprints through optimized logistics, adopting eco-friendly packaging materials, implementing ethical AI practices in customer service (Moor, 1985), and enhancing resource efficiency in digital operations (Alerasoul et al., 2022; Aziz et al., 2024; Cheng et al., 2023). While prior research has highlighted the potential of digital technologies to drive innovation and sustainability in retail settings (Ardito, 2023), the specific role of digital employees in fostering sustainable innovation within consumer-oriented businesses remains underexplored. This gap is particularly salient given the unique characteristics of digital employees as both tools for enhancing service delivery and quasi-team members that can augment human capabilities in novel ways to improve customer interactions (Van Doorn et al., 2023). The ethical concerns and anxieties surrounding AI adoption in customer-facing roles add another layer of complexity to understanding how digital employees influence organizational outcomes and consumer perceptions in retail and service contexts (Xie et al., 2024).

The current landscape of research on digital employees and sustainable innovation in retail and consumer services reveals both significant advancements and critical gaps. Digital employees use digital technology, encompassing AI chatbots, virtual assistants, and conversational agents that utilize natural language processing, machine learning algorithms, and big data analytics to interact with customers, provide personalized services, and handle inquiries in real-time (Le et al., 2024). For instance, TikTok online retailers have integrated AI chatbots into their customer service operations, enabling efficient handling of high volumes of customer interactions (Bilal et al., 2024). These chatbots can process customer queries, provide product recommendations, and resolve issues without human intervention, enhancing operational efficiency and customer satisfaction. Additionally, digital employees include technologies such as computer vision for product recognition, recommendation engines that analyze customer preferences, and sentiment analysis tools that gauge customer emotions and feedback (Huang and Gursoy, 2024). For example, AI-powered recommendation systems on TikTok suggest products to customers based on their browsing history and engagement patterns, thereby personalizing the shopping experience and increasing engagement (Van Doorn et al., 2023). While studies have highlighted the potential of digital technologies to drive innovation and sustainability in retail (Ardito, 2023), the specific role of digital employees in fostering sustainable innovation within consumer-oriented businesses remains underexplored. Extant literature has primarily focused on the direct effects of digital transformation on retail firm performance (Wang and Esperança, 2023) or the impact of specific digital technologies on innovation outcomes in consumer services (Singh et al., 2024). However, these studies have not fully captured the complex mechanisms through which digital employee adoption influences sustainable innovation performance in retail and service contexts.

The ethical concerns and anxieties surrounding AI adoption in customer-facing roles add another layer of complexity to understanding this relationship in retail and service settings (Li and Huang, 2020). The potential moderating role of ethical considerations in the adoption-innovation link has received limited attention, particularly in consumer-oriented industries where trust and human interaction are often paramount. While research has examined the mediating effects of various organizational capabilities in the digital transformation-performance relationship for retailers, the specific mediating role of ambidextrous green innovation in the context of digital employee adoption in consumer services remains unexplored. To address these research gaps, our study is guided by the following research questions.

1. How does digital employee adoption influence exploitative and exploratory green innovation in online retail settings?
2. What is the mediating role of ambidextrous green innovation (both exploitative and exploratory) in the relationship between digital employee adoption and future sustainable innovation performance?
3. How does digital employee ethical anxiety moderate the relationships between digital employee adoption, ambidextrous green innovation, and future sustainable innovation performance in online retail?

By seeking answers to these questions, we aim to contribute to the body of knowledge in several ways. First, we extend the literature on digital transformation in retail by elucidating the mechanisms through which digital employee adoption influences both exploitative and exploratory green innovation, thereby enhancing future sustainable innovation performance (Singh et al., 2024; Ardito, 2023). Second, we integrate ethical considerations into the discussion by examining the moderating effect of digital employee ethical anxiety, offering new insights into how ethical concerns impact innovation processes and outcomes in online retail (Li and Huang, 2020; Xie et al., 2024). Third, we propose and empirically validate the Adoption-Innovations-Future Performance (AIF) model, which extends the Antecedents-Behavior-Consequences (ABC) framework to capture the complex interplay between digital employee adoption, ambidextrous green innovation, ethical anxiety, and sustainable innovation performance in the context of online retailing (Aziz et al., 2024; Mehmood et al., 2024a,b; Melanson and Fahmie, 2023; Vishwakarma et al., 2023).

The novelty of our study lies in its comprehensive examination of the roles of digital employees and ethical anxiety in driving sustainable innovation within online retail contexts (Gkinko and Elbanna, 2023; Li and Huang, 2020; Yin et al., 2024). While prior research has explored the impact of digital technologies on innovation, the specific influence of digital employees combined with ethical considerations remains underexplored (Van Doorn et al., 2023; Le et al., 2024). Our study fills this gap by providing empirical evidence and theoretical advancements that offer practical implications for online retailers seeking to leverage digital employees for sustainability.

Practically, this study provides valuable insights for managers and practitioners in retail and consumer services navigating digital transformation and sustainability initiatives. By elucidating the conditions under which digital employees can foster ambidextrous green innovation in consumer-oriented contexts, this research offers actionable strategies for firms looking to enhance their sustainable innovation performance while improving customer experiences. The identification of ethical anxiety as a key moderating variable also provides practical guidance for retailers and service providers to address and mitigate the potential negative impacts of digital employee adoption on consumer trust and engagement, thereby ensuring a smoother integration of AI technologies in customer-facing roles (Le et al., 2024).

Moreover, this study's methodological approach, utilizing a two-wave time-lagged survey of online retailers and Partial Least Squares Structural Equation Modeling (PLS-SEM) analysis, contributes to the robustness of the findings and offers a reliable framework for future research in retail and consumer services. The incorporation of fuzzy-set Qualitative Comparative Analysis (fsQCA) provides a deeper understanding of the configurations leading to high sustainable innovation performance in consumer-oriented industries, further enhancing the study's practical relevance for retail and service managers (Cheng et al., 2023; Rauter et al., 2023).

This paper is organized as follows. In Section 2, we review the literature on digital employee adoption, ambidextrous green innovation, and sustainable innovation performance in retail and consumer services, identifying key research gaps, and we develop the theoretical framework and hypotheses based on the Antecedents-Behaviour-Consequences (ABC) model, with a focus on consumer-oriented contexts. In Section 3, we developed hypotheses. In Section 4, we detail the

research design, data collection from online retailers, and analytical approach. Following this, Sections 5 and 6 present the results of the empirical analysis and interviews with retail managers. Finally, in Sections 7 and 8, we discuss the theoretical and practical implications for retail and consumer services, limitations, and future research directions in this rapidly evolving field.

2. Theoretical background

A review of existing literature is essential to identify research gaps, establish theoretical foundations, and position our study within the broader academic discourse on digital employee adoption and sustainable innovation in retail and consumer services. This literature review serves three key purposes: (1) to synthesize current knowledge on future sustainable innovation performance in retail settings, particularly in relation to digital transformation; (2) to identify critical gaps in understanding the role of digital employees in driving sustainable innovation; and (3) to establish clear construct definitions that will guide our empirical investigation. The following sections examine previous studies, highlight research gaps, and define key constructs that form the foundation of our theoretical framework.

2.1. Previous studies on future sustainable innovation performance in retail and consumer services

Research on future sustainable innovation performance in retail and consumer services has evolved significantly, exploring various factors driving sustainability within organizations. Early studies (Table 1) focused on how digital technologies enable sustainable practices in retail settings (Vhatkar et al., 2024). These highlighted how digital platforms enhance organizational capabilities to adopt sustainability-oriented innovations in customer-facing operations (Ardito, 2023).

Subsequent research investigated how organizational capabilities, particularly ambidextrous innovation, mediate between digital transformation and sustainable outcomes in retail and service contexts (Sun et al., 2024). These studies revealed how retailers balance incremental and radical innovations to achieve long-term sustainability goals while meeting changing consumer demands. The integration of ambidextrous strategies proved crucial for enhancing both environmental and economic performance in retail practices (Rauter et al., 2023; Wang et al., 2023).

Another key area explored the moderating role of environmental and organizational factors in adopting digital technologies for sustainable innovation in consumer services. For instance, Singh et al. (2024) studied how ethical dilemmas and market dynamism influence digital transformation's effectiveness in driving sustainable outcomes in retail. Zhu et al. (2024) investigated AI ethical anxiety, perceived ethical risks, and ethical awareness among university students using generative AI products. Their findings underscore the significance of ethical considerations in AI adoption, highlighting how ethical anxiety can influence user engagement with AI technologies. This perspective enriches our understanding of the ethical challenges associated with digital employee adoption in retail and consumer services. These findings underscore the complexity of integrating digital technologies into customer-centric practices, where ethical concerns and changing consumer preferences can alter technology impacts.

The integration of AI technologies in retail has also been explored in the context of enhancing customer experience post-crisis recovery, highlighting the development of techno-business platforms that leverage AI for improved B2B interactions (Behera et al., 2022). Explainable AI has been identified as a crucial factor in fostering sustainable growth among consumer packaged goods retailers, enabling better decision-making and competitive advantage (Behera et al., 2023). AI-driven e-marketing strategies have been shown to transform customer engagement in the Retail 4.0 era, emphasizing the importance

Table 1
Previous studies on Future Sustainable Innovation Performance.

Author	Year	Findings	Gaps
Singh et al.	2024	Examines the impact of generative AI adoption on future organizational performance, highlighting the mediating role of explorative and exploitative innovation and the moderating role of ethical dilemmas and environmental dynamism.	Did not specifically focus on digital employees in retail and consumer services; lacked examination of ethical anxiety as a moderating factor; did not address ambidextrous green innovation in the retail context.
Rauter et al.	2023	Demonstrate that formal and social organizational controls can significantly enhance sustainability innovation performance, particularly through clear sustainability-related goals and shared values.	Did not investigate the impact of digital employee adoption; did not explore the moderating effect of ethical anxiety; lacked focus on ambidextrous green innovation in retail and consumer services.
Ardito	2023	Investigates the relationship between firm digitalization and sustainable innovation, showing that digitalization and sustainability practices are often treated independently but have significant effects on innovation outcomes.	Did not examine the specific role of digital employees in sustainable innovation; and did not consider the mediating role of ambidextrous green innovation; did not address ethical anxiety as a moderating factor.
Cheng et al.	2023	Explores how different configurations of digital technology adoption influence firms' sustainable innovation performance, identifying specific patterns that lead to high performance in Chinese high-tech firms.	Focused on manufacturing firms rather than retail and consumer services; did not consider digital employee adoption; did not address ethical anxiety or ambidextrous green innovation.
Alerasoul et al.	2022	Analyzes the synergistic impact of market and technology orientations on sustainable innovation performance, highlighting the importance of balancing these orientations in manufacturing firms.	Did not examine digital employee adoption or its effects; lacked focus on ethical anxiety and ambidextrous green innovation; studied manufacturing sector, not retail or consumer services.
Wang et al.	2023	Show that digital platform capabilities enhance sustainability-oriented innovation through open innovation, with balanced inbound and outbound innovation being crucial for improved performance.	Did not address the role of digital employees; did not consider ethical anxiety as a moderating factor; lacked focus on ambidextrous green innovation in the context of retail and consumer services.
Aziz et al.	2024	Provides a comprehensive review of Green Innovation Performance (GIP) literature, identifying key trends and research gaps in the context of sustainable product and process innovation.	Offered a broad review without empirical examination; did not focus on digital employee adoption or ethical anxiety; lacked specific insights into ambidextrous green innovation in retail and consumer services sectors.
Taghizadeh et al.	2023	Investigates the role of dynamic capabilities in triggering sustainable innovation performance in SMEs, emphasizing the importance of integrating environmental contingencies.	Focused on SMEs without addressing digital employee adoption; did not consider ethical anxiety; did not explore ambidextrous green innovation or its impact on future sustainable

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Table 1 (continued)

Author	Year	Findings	Gaps
Calik	2024	Develops the first validated scale for measuring sustainable product innovation performance, providing a tool for assessing progress in manufacturing companies.	innovation performance in retail settings. Concentrated on scale development in manufacturing context; did not examine digital employees or ethical anxiety; did not address ambidextrous green innovation in retail and consumer services industries.
Fawad Sharif et al.	2022	Examines the impact of knowledge coupling on SME innovation performance during the pandemic, identifying ambidexterity and market capitalizing agility as key mediators.	Did not focus on digital employee adoption or ethical anxiety; research centered on SMEs during the pandemic; lacked exploration of ambidextrous green innovation in the retail and consumer services context.
Lin & Mao	2023	Analyzes the mechanisms through which digital transformation affects sustainable innovation performance, focusing on the roles of digital technology-business alignment and environmental uncertainty.	Did not specifically investigate digital employee adoption; did not consider ethical anxiety; did not examine ambidextrous green innovation in retail and consumer services sectors.
Arshad et al.	2024	Explores how organizational readiness for big data analytics influences sustainable innovation performance, emphasizing the need for organizational alignment.	Focused on big data analytics rather than digital employee adoption; did not address ethical anxiety or ambidextrous green innovation; lacked application to retail and consumer services contexts.
Lee & Roh	2023	Investigate the role of coopetition strategy and digitalization capability in enhancing sustainable performance, with open innovation serving as a key mediator.	Did not examine digital employee adoption or ethical anxiety; research centered on coopetition strategy; did not explore ambidextrous green innovation in the retail and consumer services industry.
Chen et al.	2024	Studies the impact of digital transformation on sustainable business performance, highlighting the role of digital leadership and market-driven business model innovation.	Did not specifically focus on digital employee adoption; did not consider ethical anxiety; did not address ambidextrous green innovation or its influence on future sustainable innovation performance in retail settings.
Al Halbusi et al.	2024	Explores the relationship between green entrepreneurial orientations, green knowledge management, and digital transformation in driving sustainable innovation and circular economy practices.	Did not examine the role of digital employees; lacked focus on ethical anxiety as a moderating factor; did not study ambidextrous green innovation in the context of retail and consumer services.

of AI in enhancing e-retailer performance (Behera et al., 2024). The empowerment of service co-creation through AI has been investigated, revealing how AI adoption enhances collaborative value creation between customers and service providers (Behera et al., 2024). A comprehensive bibliometric review of sustainable retailing highlights its evolution and significant influence on consumer behavior, underlining

the necessity for retailers to integrate sustainability into their business models (Singh et al., 2024). Research also examined configurational aspects of digital adoption, exploring how different combinations of digital tools and strategies contribute to sustainable innovation in retail and services (Cheng et al., 2023). These studies emphasize aligning digital and sustainability strategies to achieve optimal innovation performance in consumer-facing operations.

Previous studies on innovation in online retailing have predominantly focused on technological advancements that enhance operational efficiency, improve customer experience, and strengthen competitive positioning (Vhatkar et al., 2024). While these innovations contribute to the growth and performance of online retailers, they may not necessarily address environmental or social concerns (Wang et al., 2023). In contrast, sustainable innovation in online retailing emphasizes the integration of sustainability principles into the core innovation processes (Ardito, 2023). This involves rethinking business models and practices to achieve a balance between economic objectives and the imperatives of environmental conservation and social equity (Taghizadeh et al., 2023). For instance, sustainable innovation may involve developing platforms that promote the circular economy, employing AI to minimize waste in supply chain management, or using digital employees to support ethical decision-making in customer interactions (Cheng et al., 2023; Wang and Zhang, 2024).

While advancing our understanding of sustainable innovation drivers in retail and services, gaps remain regarding digital employees' nuanced roles and associated ethical anxieties (Kumar et al., 2024). Existing literature has yet to fully explore how AI-powered virtual agents, as both tools and quasi-team members, mediate and moderate digital transformation's impact on sustainable innovation outcomes in customer service contexts. The intersection of digital employee adoption and ethical anxiety in retail settings presents a critical area that could reveal new pathways to future sustainable innovation performance in consumer services.

2.2. Gaps, questions and contributions

While existing literature has advanced our understanding of digital transformation's role in enhancing retail and service outcomes, critical gaps remain regarding digital employees' nuanced influence on sustainable innovation in consumer-facing contexts. Studies have explored digital transformation's impact on firm performance, but comprehensive research specifically addressing how AI-powered virtual agents mediate and moderate this relationship in retail settings is lacking (Singh et al., 2024). Current research focuses on broader digital strategies' direct effects on innovation outcomes, often overlooking specific mechanisms through which digital employees contribute to ambidextrous green innovation in sustainability-oriented retail and service contexts. By distinguishing sustainable innovation from general innovation, our study underscores the importance of innovations that are not only technologically advanced but also environmentally friendly and socially responsible (Arshad et al., 2024; Rauter et al., 2023; Aziz et al., 2024). This distinction is pivotal as it directly relates to how online retailers can leverage artificial intelligence digital employees to achieve sustainability goals while maintaining competitive advantage.

Moreover, ethical concerns associated with AI adoption in customer-facing roles, particularly digital employee ethical anxiety, are underexplored in moderating relationships between adoption and sustainable innovation performance in retail (Li and Huang, 2020). This study identifies a critical gap in understanding how ethical anxiety potentially inhibits yet paradoxically strengthens digital employee adoption's positive impact on future sustainable innovation performance in consumer services. This duality presents a complex interaction largely neglected in prior retail research, which often treats ethical concerns as barriers rather than multifaceted factors influencing organizational outcomes.

Another significant gap lies in the configurational approach to digital technology adoption and its impact on sustainable innovation

performance in retail and consumer services. Although recent studies have examined configurational effects of digital adoption, including various tools and strategies, understanding specific configurations leading to high sustainable innovation performance when digital employees are integrated into customer-facing workflows remains necessary (Cheng et al., 2023). This research addresses this gap by employing fuzzy-set Qualitative Comparative Analysis (fsQCA) to identify configurations of digital employee adoption, ethical anxiety, and ambidextrous green innovation resulting in optimal sustainable innovation outcomes in retail and service contexts.

This study contributes to retail and consumer services literature by developing and empirically testing the Adoption-Innovations-Future Performance model, integrating underexplored factors into a comprehensive framework. By extending the Antecedents-Behaviour-Consequences (ABC) model to include ethical anxiety as a moderating factor, this research offers a novel theoretical perspective on the complex interplay between digital employee adoption and sustainable innovation performance in customer-centric environments. The findings provide actionable insights for retail practitioners and policymakers, highlighting the importance of strategically managing ethical concerns to harness digital employees' full potential in driving sustainable innovation in consumer-facing operations.

2.3. Research construct definitions

The constructs selected for this study—digital employee adoption, ambidextrous green innovation (exploitative and exploratory), future sustainable innovation performance, and digital employee ethical anxiety—are grounded in the theoretical framework of the Adoption-Innovations-Future Performance (AIF) model (Nijhof and Rietdijk, 1999; Mehmood et al., 2024a,b; Vishwakarma et al., 2023; Melanson and Fahmie, 2023; Singh et al., 2024; Rauter et al., 2023; Li and Huang, 2020). Digital employee adoption serves as a critical antecedent, reflecting the integration of AI-powered virtual agents into organizational workflows to enhance efficiency and innovation (Singh et al., 2024; Le et al., 2024). Ambidextrous green innovation captures the dual dimensions of exploitative and exploratory innovation, which are essential for balancing incremental improvements with radical advancements in sustainability (Asiaei et al., 2023; Wang et al., 2020). Future sustainable innovation performance represents the long-term impact of innovation efforts on environmental and social outcomes, aligning with sustainability goals (Rauter et al., 2023; Ardito, 2023). Finally, digital employee ethical anxiety is included as a moderating factor, reflecting the ethical concerns and apprehensions associated with AI adoption in organizational settings (Li and Huang, 2020; Jobin et al., 2019). These constructs collectively provide a framework for examining the interplay between digital transformation, innovation processes, and sustainability outcomes. Table 2 presents the definitions of the main constructs used in this study, along with their sources in the literature.

Digital employee adoption refers to the extent to which an organization incorporates AI-powered virtual agents into its operations, emphasizing their role in enhancing efficiency and promoting collaboration with human employees (Singh et al., 2024). This definition extends beyond mere technology implementation to capture the integration of digital employees as quasi-team members that augment human capabilities (Huang and Gursoy, 2024).

Ambidextrous green innovation refers to an organization's ability to simultaneously pursue both exploitative and exploratory green innovation activities (Asiaei et al., 2023; Wang et al., 2020). Exploitative green innovation focuses on the refinement and enhancement of existing environmentally friendly products, services, and processes to improve ecological performance within current markets (Asiaei et al., 2023). This involves incremental improvements and optimization of current sustainability practices. Exploratory green innovation, in contrast, involves the development and adoption of entirely new green products, services, and processes that open up new markets and significantly advance

Table 2
Research construct definitions.

Construct	Definition	Sources
Digital Employee Adoption	The extent to which an organization incorporates digital employees into its operations, emphasizing their role in enhancing efficiency, and promoting collaboration.	Singh et al. (2024); Wang and Esperança (2023)
Exploitative Green Innovation	The process of improving and optimizing existing green products, processes, and technologies to enhance environmental performance within current markets.	Asiaei et al. (2023); Wang et al. (2020)
Exploratory Green Innovation	The development and adoption of novel green products, processes, and technologies aimed at entering new markets and advancing environmental sustainability.	Asiaei et al. (2023); Wang et al. (2020)
Future Sustainable Innovation Performance	The anticipated capability of a firm to improve environmental and social outcomes through innovations in product design, process efficiency, pollution reduction, and social responsibility over the next three years.	Rauter et al. (2023)
Digital Employee Ethical Anxiety	The apprehension and concerns within an organization regarding the ethical implications, privacy issues, job displacement, and transparency associated with the adoption of digital employees.	Jobin et al. (2019); Li and Huang (2020)

environmental sustainability (Wang et al., 2020). This includes radical innovations and pioneering solutions to environmental challenges. By balancing these two approaches, organizations can effectively enhance their overall sustainable innovation performance and remain competitive in rapidly evolving markets (Xie et al., 2024).

Future sustainable innovation performance represents a firm's anticipated capability to improve environmental and social outcomes through innovations in product design, process efficiency, pollution reduction, and social responsibility over the next three years (Rauter et al., 2023). This forward-looking construct emphasizes the long-term impact of current innovation initiatives on sustainability goals (Ardito, 2023).

Digital employee ethical anxiety refers to the apprehension and concerns within an organization regarding the ethical implications, privacy issues, job displacement, and transparency associated with the adoption of digital employees (Mason, 1986; Li and Huang, 2020). This construct captures the psychological and social dimensions of integrating AI technologies in the workplace (Jobin et al., 2019).

These construct definitions provide a foundation for examining the complex relationships between digital employee adoption, ambidextrous green innovation, and sustainable innovation performance. The next section develops the theoretical framework that guides this investigation.

2.4. Antecedents-Behaviour-Consequences (ABC) model

The Antecedents-Behaviour-Consequences (ABC) model is a theoretical framework originally developed to understand and predict human behavior by examining the interplay between antecedent conditions, behavior, and subsequent consequences (Nijhof and Rietdijk, 1999). The model posits that behavior is a function of its antecedents and consequences, where antecedents are the stimuli or events that trigger a particular behavior, and consequences are the outcomes that follow the behavior, influencing its future occurrence (Mehmood et al., 2024a,b). Technology Acceptance Model emphasizes perceived usefulness and perceived ease of use as key factors influencing technology

adoption (Davis, 1989). Diffusion of Innovations Theory highlights factors that influence adoption rates, such as relative advantage, compatibility, complexity, trialability, and observability, and emphasizes the role of opinion leaders in facilitating this process (Rogers et al., 2014). These mature theories provide inspiration for us to expand the ABC model. In organizational contexts, the ABC model has been widely employed to analyze how various factors—such as environmental conditions, organizational culture, or individual psychological states—affect employee behavior and, ultimately, organizational outcomes (Vishwakarma et al., 2023). By establishing a causal relationship between these elements, the ABC model provides a structured approach to identifying the drivers of specific behaviors and the impact of those behaviors on organizational performance (Melanson and Fahmie, 2023).

This study applies the ABC model to explore the impact of digital employee adoption on future sustainable innovation performance, mediated by ambidextrous green innovation and moderated by digital employee ethical anxiety. The antecedent in this study is the adoption of digital employees, which triggers specific organizational behaviors in the form of exploitative and exploratory green innovations. These behaviors, in turn, lead to various consequences, notably future sustainable innovation performance. The ABC model is particularly relevant here because it allows for the systematic examination of how the introduction of digital employees—a significant antecedent—shapes green innovation strategies within firms and how these strategies influence sustainability outcomes (Aziz et al., 2024). The inclusion of digital employee ethical anxiety as a moderating variable adds complexity to the model by highlighting how ethical concerns can alter the relationship between digital employee adoption and green innovation behavior, thereby affecting the overall consequences on sustainable innovation (Jobin et al., 2019). This approach not only extends the ABC model's applicability to the digital transformation context but also enriches our understanding of how digital and ethical dynamics interact to shape organizational performance (Dzandu, 2023).

2.5. Adoption-Innovations-Future Performance (AIF) model

The Adoption-Innovations-Future Performance Model builds upon and extends the ABC framework to elucidate the complex relationships between digital employee adoption, ambidextrous green innovation, and future sustainable innovation performance. This model posits that the adoption of digital employees serves as a critical antecedent that triggers both exploitative and exploratory green innovation behaviors within organizations (Singh et al., 2024). These innovation behaviors, in turn, lead to enhanced future sustainable innovation performance as a key consequence (Rauter et al., 2023). The AIF Model incorporates digital employee ethical anxiety as a moderating factor, recognizing that ethical concerns can significantly influence the relationship between adoption and innovation outcomes (Li and Huang, 2020). By integrating these elements, the AIF Model provides a comprehensive framework for understanding how organizations can leverage digital technologies to drive sustainable innovation while navigating ethical challenges. This model is particularly well-suited for the present study as it captures the nuanced interplay between technological adoption, innovation processes, and sustainability outcomes in the context of digital transformation (Ardito, 2023). The inclusion of ethical considerations aligns with the growing emphasis on responsible AI adoption in organizational settings (Jobin et al., 2019). The AIF Model's holistic approach enables a more nuanced examination of the mechanisms through which digital employee adoption influences sustainable innovation performance, thereby advancing both theoretical understanding and practical insights in this critical domain.

This theoretical framework is particularly suited to this study's focus on digital employees' impact on sustainable innovation performance, as it encapsulates the multifaceted interactions between digital transformation, organizational capabilities, and ethical considerations. The AIF model's relevance is further underscored by its ability to account for

the dual role of digital employees as both operational tools and quasi-team members, a concept that aligns with emerging research on digital transformation (Le et al., 2024). By incorporating ambidextrous green innovation into the framework, the model addresses the need for a balanced approach to innovation, where firms must simultaneously optimize existing green technologies while exploring new, sustainable solutions (Asiaei et al., 2023). Moreover, the model's emphasis on ethical anxiety acknowledges the growing importance of ethical considerations in AI adoption, providing a comprehensive perspective on how these concerns can either hinder or enhance the innovation process (Jobin et al., 2019). The development of the AIF model, is thus well-aligned with the objectives of this research. It not only provides a theoretical basis for the hypotheses but also serves as a guide for the empirical analysis, ensuring that the study's findings are grounded in a rigorous conceptual framework. By applying the AIF model, this research contributes to the broader discourse on digital transformation and sustainable innovation, offering new insights into how firms can strategically manage digital employees to achieve long-term sustainability goals.

2.6. Control variables and research model

The research model in this study is grounded in the Antecedents-Behavior-Consequences (ABC) framework and the proposed Adoption-Innovations-Future Performance model. To enhance the robustness of the findings, several control variables were incorporated based on their potential influence on sustainable innovation performance. Firm size and age were included as controls, as larger and more established firms may possess greater resources and capabilities for innovation (Wang et al., 2023). The respondent's position within the organization was also controlled for, as the managerial level may affect perceptions of digital employee adoption and innovation outcomes (Singh et al., 2024). Industry category was included to account for potential sector-specific effects on sustainable innovation (Asiaei et al., 2023). The inclusion of these control variables allows for a more precise examination of the relationships between the focal constructs while accounting for potential confounding factors. By controlling for these variables, the study aims to isolate the effects of digital employee adoption, ambidextrous green innovation, and ethical anxiety on future sustainable innovation performance (Rauter et al., 2023). This approach aligns with best practices in management research and enhances the validity of the findings. Fig. 1 presents the conceptual model, which integrates the hypothesized relationships among the key constructs.

3. Hypotheses development

3.1. Adoption-innovations (A-I)

The integration of digital employees into retail and service organizations plays a pivotal role in enhancing exploitative green innovation by specifically influencing organizational routines, activities, decisions, and operations. Digital employees, equipped with advanced data analytics capabilities, enable precise optimization of existing green processes (Asiaei et al., 2023). For instance, they analyze vast amounts of customer and operational data to identify inefficiencies in resource utilization, leading to incremental improvements in environmental performance (Wang et al., 2020). By automating routine customer service tasks, digital employees free up human employees to concentrate on refining current sustainable practices (Singh et al., 2024). This shift allows personnel to engage in creative problem-solving and fine-tuning of existing green products and services, thereby enhancing the overall retail experience. Moreover, the continuous learning capacity of digital employees facilitates the ongoing improvement of established green technologies by incorporating real-time feedback and best practices (Ardito, 2023). These AI agents can monitor sustainability metrics and suggest adjustments to operations, ensuring that organizations

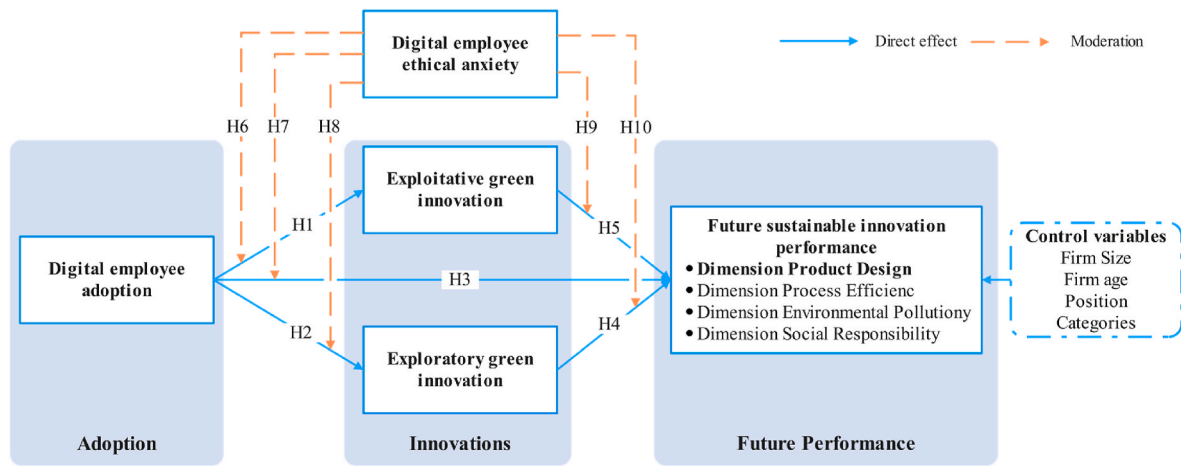


Fig. 1. Conceptual model: Adoption-Innovations-Future Performance (AIF) Model.

systematically strengthen their existing green market positions (Cheng et al., 2023). Consequently, by directly affecting daily routines and strategic decisions, digital employee adoption positively influences exploitative green innovation within retail and service settings. Based on these arguments, we propose.

H1. Digital employee adoption positively influences exploitative green innovation in retail and service settings.

Beyond optimizing existing processes, digital employee adoption significantly contributes to exploratory green innovation by transforming activities, routines, and decision-making processes within organizations. Digital employees' advanced pattern recognition capabilities enable the identification of new green market opportunities and emerging sustainability trends that may be overlooked by human employees (Wang et al., 2023). By processing and analyzing vast amounts of unstructured consumer data from diverse sources, these AI agents generate novel insights that inform strategic decisions about new product development and market entry. Additionally, digital employees facilitate rapid prototyping and simulation through AI-powered capabilities that support radical sustainable innovation experiments (Ardito, 2023). This allows organizations to experiment with innovative eco-friendly solutions more efficiently and cost-effectively. By automating routine tasks, digital employees enable human workers to engage in more creative and exploratory activities, fostering a culture of experimentation and risk-taking essential for breakthrough discoveries in sustainability. Their cognitive abilities facilitate access to cutting-edge knowledge on the latest sustainability practices and technologies, thus enhancing exploratory innovation efforts (Asiaei et al., 2023). The synergy between human creativity and AI-powered analytical and learning capabilities creates an ideal environment for developing revolutionary eco-friendly technologies that resonate with environmentally conscious consumers (Cheng et al., 2023). Therefore, by influencing activities and decision-making processes, digital employee adoption positively impacts exploratory green innovation in retail and service organizations. Based on these arguments, we propose.

H2. Digital employee adoption positively influences exploratory green innovation in retail and service settings.

Moreover, digital employee adoption directly enhances a retail organization's future sustainable innovation performance by impacting strategic routines, decisions, and operations. Real-time monitoring capabilities of digital employees enable dynamic adjustments to sustainability initiatives, allowing firms to respond promptly to changes in environmental regulations and market demands (Bilal et al., 2024). Predictive analytics provided by these AI agents support long-term sustainability planning by forecasting trends in green consumer

behavior and resource availability (Wang et al., 2023). The integration capabilities of digital employees enhance cross-functional sustainability coordination by facilitating effective collaboration and knowledge sharing across different departments and external partners in the retail value chain (Al Halbusi et al., 2024). This holistic approach breaks down silos that often hinder sustainable innovation efforts, leading to improved product design, service efficiency, and social responsibility. Additionally, by automating routine customer service tasks, digital employees allow human employees to focus on more strategic and creative sustainability-focused activities, fostering a culture of innovation essential for long-term performance. This direct influence on routines and strategic decisions positions retail firms to enhance their future sustainable innovation performance, aligning with evolving consumer preferences and stakeholder expectations. Based on these arguments and the theoretical underpinnings of the AIF Model in the context of retail and consumer services, we propose.

H3. Digital employee adoption positively influences future sustainable innovation performance in retail and service organizations.

3.2. Innovations-future performance (I-F)

Exploratory green innovation plays a crucial role in driving future sustainable innovation performance in retail and consumer services. The Adoption-Innovations-Future Performance Model posits that innovation behaviors, particularly exploratory innovations, serve as key mediators between digital adoption and future performance outcomes in service industries (Singh et al., 2024). In the retail context, exploratory green innovation involves developing novel eco-friendly products, processes, and technologies aimed at creating new sustainable service offerings and advancing environmental sustainability (Asiaei et al., 2023). This type of innovation enables retailers to discover untapped opportunities in the sustainability domain, potentially leading to breakthrough solutions that can enhance future sustainable innovation performance. By engaging in exploratory green innovation, retail organizations can develop radically new eco-friendly technologies, products, and service models that address emerging environmental challenges and evolving consumer needs (Sun et al., 2024). Such innovations often involve higher degrees of novelty and experimentation in service delivery, which can lead to more substantial improvements in sustainability outcomes compared to incremental innovations in traditional retail practices (Ardito, 2023). Exploratory green innovation allows retailers to stay ahead of evolving sustainability regulations and changing consumer expectations, positioning them favorably for future sustainable performance in the competitive retail landscape (Wang et al., 2023). The process of exploring new green technologies and sustainable service markets also enhances a retailer's adaptive capabilities, making it more

resilient and better equipped to navigate future sustainability challenges in the rapidly changing retail environment (Taghizadeh et al., 2023). Successful exploratory green innovations can establish a retailer as a sustainability leader, attracting environmentally conscious consumers and investors, which further contributes to long-term sustainable performance in the retail sector (Gupta and Rathore, 2024). Given these arguments and the theoretical underpinnings of the AIF Model in the retail context, the following hypothesis is proposed.

H4. Exploratory green innovation positively influences future sustainable innovation performance in retail and consumer services.

Exploitative green innovation plays a crucial role in enhancing future sustainable innovation performance in retail and consumer services. The Adoption-Innovations-Future Performance Model posits that innovation behaviors, particularly exploitative innovations, serve as key mediators between digital adoption and future performance outcomes in service industries (Singh et al., 2024). In the retail context, exploitative green innovation involves refining and extending existing sustainability-related knowledge, skills, and processes to improve current green products, services, and technologies (Asiaei et al., 2023). This type of innovation enables retailers to incrementally enhance their environmental performance within established markets, leading to more efficient resource utilization and reduced environmental impact in their service operations (Wang et al., 2020). By focusing on exploitative green innovation, retail organizations can systematically optimize their existing green technologies and service processes, resulting in immediate and tangible improvements in sustainability metrics relevant to the retail sector (Rauter et al., 2023). Exploitative green innovation allows retailers to respond more effectively to current sustainability regulations and consumer expectations, thereby enhancing their short-term environmental performance in service delivery (Ardito, 2023). The continuous refinement of existing green products, services, and processes also contributes to cost reduction and efficiency gains in retail operations, which are critical for maintaining competitiveness while pursuing sustainability goals in the highly competitive retail environment (Alerasoul et al., 2022). Exploitative green innovation fosters a culture of continuous improvement in sustainability practices within retail organizations, encouraging employees to identify and implement incremental enhancements in their day-to-day service operations (Cheng et al., 2023). This ongoing focus on optimizing current green technologies and service processes creates a solid foundation for long-term sustainable innovation performance by ensuring that existing sustainability initiatives in retail are continually refined and improved. Given these arguments and the theoretical underpinnings of the AIF Model in the retail context, the following hypothesis is proposed.

H5. Exploitative green innovation positively influences future sustainable innovation performance in retail and consumer services.

3.3. Digital employee ethical anxiety as moderator

Digital employee ethical anxiety plays a crucial moderating role in the relationship between digital employee adoption and exploitative green innovation in retail and consumer services. The Adoption-Innovations-Future Performance Model posits that the adoption of digital technologies serves as a key antecedent to innovation behaviors within retail organizations (Singh et al., 2024). However, the ethical concerns surrounding AI adoption can influence this relationship in the service context. Digital employee ethical anxiety refers to the apprehension and concerns within a retail organization regarding the ethical implications, privacy issues, job displacement, and transparency associated with the adoption of digital employees in customer-facing roles (Li and Huang, 2020). As retailers integrate AI-powered digital employees into their service operations, they may encounter resistance and anxiety from human employees, potentially hindering the positive effects of adoption on innovation outcomes in the retail environment.

First, heightened ethical concerns may lead to reduced trust and acceptance of digital employees among human retail workers, limiting collaborative efforts essential for exploitative innovation in service delivery (Gkinko and Elbanna, 2023). When retail employees harbor anxieties about AI ethics, they may be less inclined to share knowledge or engage in joint problem-solving with digital counterparts, impeding the refinement of existing green products and service processes. Second, ethical anxiety can create cognitive dissonance within the retail organization, diverting attention and resources away from innovation activities towards addressing ethical concerns in customer interactions (Yin et al., 2024). This shift in focus may result in decreased efficiency in leveraging digital employees for incremental improvements in sustainability practices within retail operations (Kumar et al., 2024). Ethical anxiety may lead to increased regulatory scrutiny and self-imposed restrictions on the use of digital employees in retail, potentially limiting their application in certain areas of green innovation in customer service (Jobin et al., 2019). Retail organizations grappling with ethical concerns may adopt a more cautious approach to AI integration in customer-facing roles, reducing the scope and intensity of exploitative green innovation efforts in service delivery. The perception of ethical risks associated with digital employees can create a climate of uncertainty in retail environments, potentially dampening the motivation and willingness of human employees to engage in collaborative innovation with AI systems in customer service contexts (Kong et al., 2023).

The negative moderating effect of digital employee ethical anxiety is particularly salient in the context of exploitative green innovation in retail, which relies heavily on the optimization and refinement of existing service processes and technologies. Exploitative innovation in retail often requires a deep understanding of current practices and close collaboration between human and digital employees to identify incremental improvements in customer service and sustainability (Asiaei et al., 2023). When ethical anxiety is high in retail settings, this collaborative synergy may be disrupted, reducing the effectiveness of digital employees in driving exploitative green innovation in service delivery. Empirical evidence from related fields supports the notion that ethical concerns can hinder the positive outcomes of technology adoption in service industries. For instance, studies have shown that privacy concerns negatively moderate the relationship between big data analytics adoption and firm performance in retail contexts (Arshad et al., 2024). Similarly, research on AI adoption in service industries has demonstrated that ethical anxieties can impede the realization of AI's potential benefits in improving customer experiences and operational efficiency (Zhang and Chen, 2024). Thus, the following hypothesis is proposed in the retail and consumer services context.

H6. Digital employee ethical anxiety negatively moderates the relationship between digital employee adoption and exploitative green innovation in retail and consumer services.

Digital employee ethical anxiety can moderate the relationship between digital employee adoption and future sustainable innovation performance in retail and consumer services. The Adoption-Innovations-Future Performance Model posits that the adoption of digital technologies serves as a key antecedent that can directly influence future organizational outcomes in service industries (Singh et al., 2024). However, the ethical concerns surrounding AI adoption may complicate this relationship in retail contexts. As retailers integrate AI-powered digital employees into their service operations, they may encounter resistance and anxiety from human employees, potentially hindering the positive effects of adoption on innovation outcomes in customer-facing roles (Gkinko and Elbanna, 2023). Heightened ethical concerns may lead to reduced trust and acceptance of digital employees among human retail workers, limiting collaborative efforts essential for sustainable innovation in service delivery (Kong et al., 2023). This climate of uncertainty and ethical apprehension can dampen the motivation and willingness of human employees to engage in innovative practices enabled by digital employees in retail environments (Yin et al., 2024). Retail organizations

grappling with ethical concerns may adopt a more cautious approach to AI integration in customer service, reducing the scope and intensity of sustainable innovation efforts in their service offerings (Jobin et al., 2019). Paradoxically, this heightened ethical awareness may also drive retail organizations to more carefully consider the long-term implications of their innovation practices on customer experiences and societal well-being, potentially leading to more robust and truly sustainable innovations in service delivery. This complex interplay suggests that while digital employee ethical anxiety may initially hinder the positive relationship between adoption and sustainable innovation performance in retail, it may also catalyze more thoughtful and responsible innovation practices in customer service in the long run. Based on these arguments and the theoretical underpinnings of the AIF Model in the retail context, the following hypothesis is proposed.

H7. Digital employee ethical anxiety negatively moderates digital employee adoption and future sustainable innovation performance in retail and consumer services.

Digital employee ethical anxiety can moderate the relationship between digital employee adoption and exploratory green innovation in retail and consumer services. The Adoption-Innovations-Future Performance Model posits that the adoption of digital technologies serves as a key antecedent to innovation behaviors, including exploratory innovations that involve developing novel products, processes, and technologies in service industries (Singh et al., 2024). However, ethical concerns surrounding AI adoption may complicate this relationship in the context of exploratory green innovation in retail. As retail organizations integrate AI-powered digital employees into their service operations, they may encounter resistance and anxiety from human employees, potentially hindering the positive effects of adoption on radical innovation outcomes in customer-facing roles (Gkinko and Elbanna, 2023). Heightened ethical concerns may lead to reduced trust and acceptance of digital employees among human retail workers, limiting collaborative efforts essential for exploratory innovation in service delivery (Kong et al., 2023). This climate of uncertainty and ethical apprehension can dampen the motivation and willingness of human employees to engage in risky, experimental practices enabled by digital employees in retail environments (Yin et al., 2024). Retail organizations grappling with ethical concerns may adopt a more cautious approach to AI integration in customer service, reducing the scope and intensity of exploratory green innovation efforts in their service offerings (Jobin et al., 2019). The negative moderating effect of digital employee ethical anxiety is particularly salient in the context of exploratory green innovation in retail, which relies heavily on experimentation and risk-taking to develop novel eco-friendly technologies and enter new sustainable service markets (Asiaei et al., 2023). When ethical anxiety is high in retail settings, the collaborative synergy between human creativity and AI-powered analytical capabilities may be disrupted, reducing the effectiveness of digital employees in driving exploratory green innovation in service delivery. Based on these arguments and the theoretical underpinnings of the AIF Model in the retail context, the following hypothesis is proposed.

H8. Digital employee ethical anxiety negatively moderates digital employee adoption and exploratory green innovation in retail and consumer services.

Digital employee ethical anxiety may negatively moderate the relationship between exploitative green innovation and future sustainable innovation performance in retail and consumer services. The Adoption-Innovations-Future Performance Model posits that innovation behaviors serve as key mediators between digital adoption and future performance outcomes in service industries (Singh et al., 2024). However, ethical concerns surrounding AI adoption can complicate this relationship, particularly for exploitative innovations that refine existing service processes in retail (Asiaei et al., 2023). When ethical anxiety is high in retail settings, organizations may become overly cautious in leveraging

exploitative green innovations enabled by digital employees, limiting their potential to enhance sustainable performance in service delivery (Gkinko and Elbanna, 2023). This hesitancy can manifest as the reduced willingness to fully implement or scale up incremental sustainability improvements in customer service, even when they have proven effective (Ardito, 2023). Heightened ethical anxiety may lead to increased scrutiny and bureaucratic hurdles for exploitative innovations in retail, slowing their translation into tangible sustainability outcomes in service operations (Zhang and Chen, 2024). The cognitive dissonance created by ethical concerns can divert resources and attention away from effectively capitalizing on exploitative green innovations in retail, diminishing their impact on future sustainable performance in customer-facing roles (Yin et al., 2024). Consequently, while exploitative green innovation generally contributes positively to sustainable performance in retail, this relationship may be weakened when digital employee ethical anxiety is high in service contexts. Based on these arguments and the AIF Model framework applied to retail and consumer services, the following hypothesis is proposed.

H9. Digital employee ethical anxiety negatively moderates exploitative green innovation and future sustainable innovation performance in retail and consumer services.

Digital employee ethical anxiety may negatively moderate the relationship between exploratory green innovation and future sustainable innovation performance in retail and consumer services. The Adoption-Innovations-Future Performance Model posits that innovation behaviors serve as key mediators between digital adoption and future performance outcomes in service industries (Singh et al., 2024). However, ethical concerns surrounding AI adoption can complicate this relationship, particularly for exploratory innovations that involve developing novel eco-friendly products and technologies in retail contexts (Asiaei et al., 2023). When ethical anxiety is high in retail settings, organizations may become overly cautious in pursuing radical green innovations enabled by digital employees, limiting their potential to enhance long-term sustainability performance in service delivery (Ardito, 2023). This hesitancy can manifest as the reduced willingness to invest in or scale up breakthrough sustainability solutions in customer service, even when they show promise (Cheng et al., 2023). Heightened ethical anxiety may lead to increased scrutiny and bureaucratic hurdles for exploratory innovations in retail, slowing their translation into tangible sustainability outcomes in service operations (Zhang and Chen, 2024). The cognitive dissonance created by ethical concerns can divert resources and attention away from effectively capitalizing on exploratory green innovations in retail, diminishing their impact on future sustainable performance in customer-facing roles (Yin et al., 2024). Consequently, while exploratory green innovation generally contributes positively to sustainable performance in retail, this relationship may be weakened when digital employee ethical anxiety is high in service contexts. Based on these arguments and the AIF Model framework applied to retail and consumer services, the following hypothesis is proposed.

H10. Digital employee ethical anxiety negatively moderates exploratory green innovation and future sustainable innovation performance in retail and consumer services.

4. Research design

Fig. 2 illustrates the comprehensive methodology employed in our study. It presents a sequential flow of our research process, starting with the development of the measurement instrument through an extensive literature review and expert consultations. The flowchart includes the pilot testing phase with senior executives from online retailers on TikTok, ensuring the instrument's reliability and validity. It then outlines the two-wave data collection process, highlighting measures taken to mitigate common method bias, such as time separation and anonymity assurance. The figure also depicts the data analysis techniques used,

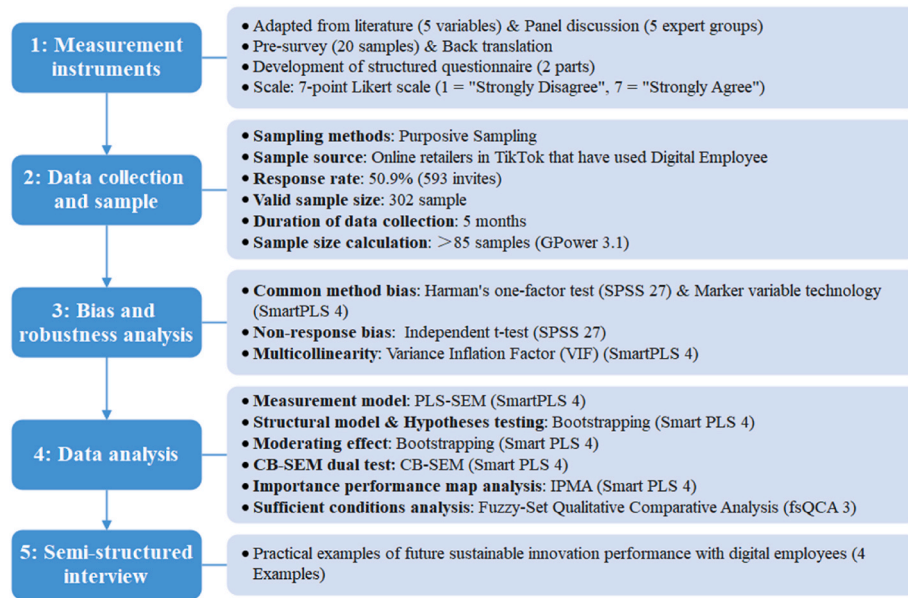


Fig. 2. Methodology flowchart.

including PLS-SEM for hypothesis testing and fsQCA for exploring complex causal relationships (Rahman et al., 2023). This visual representation provides a clear overview of our mixed-methods approach, integrating both quantitative and qualitative methodologies to enhance the robustness and credibility of our findings (Jauhari et al., 2024).

4.1. Research context

The rapid proliferation of digital technologies is fundamentally transforming business operations across industries, with digital employees emerging as a key driver of this transformation (Singh et al., 2024). This study focuses on online retailers operating on TikTok, a context that provides a unique opportunity to examine the adoption of digital employees and its impact on sustainable innovation performance. The e-commerce landscape on TikTok represents a dynamic and fast-paced environment where digital technologies play a crucial role in enhancing operational efficiency and customer engagement (Wang et al., 2023). Online retailers on TikTok face intense competition and rapidly changing consumer preferences, making innovation and sustainability critical for long-term success (Lee and Roh, 2023). These firms are at the forefront of adopting digital employees, leveraging AI-powered virtual agents to handle various tasks from customer service to content creation (Le et al., 2024). This context allows for a nuanced examination of how digital employee adoption influences both exploitative and exploratory green innovations, as well as future sustainable innovation performance (Asiaei et al., 2023).

The choice of TikTok retailers as the research sample is particularly relevant due to the platform's emphasis on short-form video content, which creates unique challenges and opportunities for sustainable innovation. These retailers must balance the need for rapid content creation and customer engagement with sustainable business practices, making them an ideal setting to study the interplay between digital transformation and sustainability (Al Halbusi et al., 2024). The global reach of TikTok enables the study to capture insights from a diverse range of markets and cultural contexts, enhancing the generalizability of the findings (Cheng et al., 2023). By focusing on this specific context, the study aims to contribute to the growing body of literature on digital transformation and sustainable innovation in the e-commerce sector. The findings from this research have the potential to inform both theoretical understanding and practical strategies for leveraging digital employees to enhance sustainable innovation performance in rapidly

evolving digital marketplaces.

4.2. Questionnaire design

The development of the measurement instrument followed a rigorous multi-stage process to ensure validity and reliability. Initially, an extensive literature review was conducted to identify relevant constructs and adapt existing scales to the context of digital employee adoption and sustainable innovation (Singh et al., 2024). The preliminary questionnaire was then refined through iterative discussions among the research team, incorporating insights from prior studies on digital transformation and sustainability (Ardito, 2023).

To enhance content validity, a panel of five experts from diverse backgrounds, including Digital business, artificial intelligence, sustainable development, was convened. Through three rounds of deliberation, the panel evaluated item relevance, clarity, and comprehensiveness (Cheng et al., 2023). This process resulted in substantial modifications to improve the instrument's theoretical grounding and practical applicability.

A pilot test was subsequently conducted with 20 senior executives from online retailers in TikTok to assess the questionnaire's psychometric properties. Feedback from respondents led to further refinements, such as rewording ambiguous items and eliminating redundancies to enhance readability (Taghizadeh et al., 2023). The pilot data demonstrated satisfactory internal consistency (Cronbach's alpha >0.7) for all constructs.

The outputs from our qualitative study, namely the insights and practical examples obtained from the semi-structured interviews, were used to enrich our discussion and interpretation of the quantitative results (Rahman et al., 2023). These qualitative insights helped to contextualize our statistical findings, offering a deeper understanding of how digital employee adoption influences sustainable innovation performance in practice (Jauhari et al., 2024). By linking the qualitative outputs to the quantitative analysis, we ensure a coherent and comprehensive narrative that connects empirical data with real-world applications. This approach enhances the relevance of our research and provides valuable implications for both scholars and practitioners. Two bilingual experts independently translated the questionnaire between English and the target language, followed by reconciliation of discrepancies. This process helped identify and resolve potential linguistic and cultural issues, enhancing the instrument's validity across

different contexts (Lee and Roh, 2023).

The final questionnaire (Table 3) employed 7-point Likert scale (1 = "strongly disagree" to 7 = "strongly agree") for most constructs, allowing for nuanced measurement of respondents' perceptions (Al Halbusi et al., 2024). This comprehensive approach to questionnaire development aimed to maximize the reliability and validity of the data collected for subsequent analysis.

4.3. Data collection and sample

To enrich our study and provide practical insights, we incorporated semi-structured interviews with senior executives from online retailers on TikTok. These interviews offered valuable real-world examples illustrating how digital employees contribute to sustainable innovation performance. While our primary approach is quantitative, the qualitative data obtained from the interviews complement and enhance our findings, providing depth and context to the statistical results. Therefore, we employ a mixed-methods approach, integrating both quantitative and qualitative data to offer a comprehensive understanding of the research problem.

Given the specific nature of our research focusing on digital employee adoption in online retail, we needed to ensure that respondents had direct experience with implementing and managing digital employees in their organizations. Purposive sampling allowed us to target senior executives, directors, and managers who were directly involved in digital transformation initiatives and could provide informed responses about their firms' innovation strategies and sustainability practices. To ensure data quality and relevance, we needed respondents from firms that had already adopted digital employees and were actively using them in their operations. Random sampling might have included firms without digital employee implementation, which would not have been suitable for our research objectives. Through purposive sampling, we could screen potential respondents based on this crucial criterion. We aimed to achieve comprehensive representation across different regions (eastern, northern, western, and southern China) and various retail categories on TikTok. Purposive sampling enabled us to strategically select participants that would provide both breadth and depth in our data collection, ensuring adequate representation across different retail segments and geographical locations.

Data were collected using an online survey platform (<https://www.wjx.cn>), which facilitated the efficient distribution and collection of responses. The survey process was conducted in two waves to mitigate common method bias. In the first wave, respondents were asked about their firm's size, age, position, and categories, as well as their perceptions of digital employee adoption, exploitative green innovation, and exploratory green innovation. The initial survey also included a screening question to ensure that only respondents from firms that had adopted digital employees participated. For exclusion, we applied these criteria: 1) Position exclusion: Employees below the managerial level were excluded as they might lack a comprehensive understanding of organizational strategies and outcomes; 2) Non-adopters exclusion: Firms that had not yet adopted digital employees were excluded through our screening question; 3) Invalid responses: Responses with incomplete data or inconsistent answers were excluded from the final sample. The first wave garnered 593 invitations, with a response rate of 50.9%, resulting in 302 valid responses. Approximately one month later, the second wave of the survey was administered. This follow-up survey focused on future sustainable innovation performance, digital employee ethical anxiety, and marker variables. To encourage participation, we sent a reminder email to those who had participated in the first wave. This two-wave approach allowed us to reduce potential biases and capture more accurate and reliable data. Data were collected using an online survey platform, which facilitated the efficient distribution and collection of responses. The survey was divided into two sections: the first gathered company information, and the second focused on the study's key variables. To ensure anonymity and confidentiality,

Table 3
Questionnaire.

Questions	References
Digital Employee Adoption (DEA)	Our firm actively incorporates digital employees into our business operations. Digital employees play a significant role in enhancing our organizational efficiency. We regularly update and improve our digital employee capabilities. Our staff is encouraged to collaborate with digital employees in their daily tasks. The adoption of digital employees is a key part of our firm's digital transformation strategy. Singh et al. (2024); Wang and Esperança (2023)
Exploitative Green Innovation (IGI)	Our firm actively improves current green products, processes, and services. Our firm actively adjusts current green products, processes, and services. Our firm actively strengthens the current green market. Our firm actively strengthens current green technology. Asiaei et al. (2023); Wang et al. (2020)
Exploratory Green Innovation (RGI)	Our firm actively adopts new green products, processes, and services. Our firm actively exploits new green products, processes, and services. Our firm actively discovers new green markets. Our firm actively enters new green technology. Rauter et al. (2023)
Future Sustainable Innovation Performance (FSIP)	Our firm is likely to develop products using more environmentally friendly materials in the next three years. We anticipate significant improvements in the recyclability of our products within the coming three-year period. Our firm expects to achieve more efficient resource utilization in our production processes over the next three years. We project a substantial reduction in our overall resource consumption within the upcoming three-year timeframe. Our firm foresees a notable decrease in environmental pollution generation in the next three years. We anticipate a significant reduction in logistics-related emissions within the coming three-year period. Our firm expects to implement measures that will greatly enhance employee health over the next three years. We project substantial improvements in our firm's social practices within the upcoming three-year timeframe.
Digital Employee Ethical Anxiety (DEEA)	Our firm is anxious that digital employees may violate privacy regulations. We are concerned that digital employees might exhibit biased behavior. Our firm worries that digital employees could replace human jobs within our organization. Jobin et al. (2019); Li and Huang (2020)

(continued on next page)

Table 3 (continued)

Questions	References
We feel uneasy about the learning process required for our staff to work effectively with digital employees.	
Our firm is anxious about the existential risks posed by the adoption of digital employees.	
We are concerned that using digital employees may go against our ethical standards.	
Our firm worries about the potential development of artificial consciousness in digital employees.	
We are apprehensive about the lack of transparency in the functioning of digital employees.	

respondents were informed that their data would be used solely for academic purposes and that their privacy would be strictly protected.

Before data collection, we used G*Power 3.1.9.7 to calculate the required sample size. Based on the recommended parameters (effect size $f^2 = 0.15$, α err prob = 0.05, power level = 0.8), the minimum sample size needed was 85. Our final sample size of 302 valid responses exceeded this requirement, ensuring sufficient statistical power for the study. The ten-times rule method indicated a minimum sample size of 50, which our sample also exceeded. The sampling formula used was $n = N/[1+N \cdot e^2]$, where n is the sample size, N is the population size, and e is the margin of error (Behera et al., 2024). With an initial pool of 593 potential respondents ($N = 593$) and using a 5% margin of error ($e = 0.05$), this yielded a required sample size of 239. Our final sample of 302 valid responses exceeded this requirement, providing a confidence level of 95%. The characteristics of the final sample are presented in Table 4.

4.4. Analytical technology

In establishing and validating the measurement and structural models, this study employed Partial Least Squares Structural Equation Modeling (PLS-SEM) using the SmartPLS 4 software. PLS-SEM is widely recognized in the management field as a robust tool for handling complex causal relationships, particularly when data do not conform to a normal distribution. Given the interdependencies among multiple variables in our study, PLS-SEM was the optimal analytical technique. This approach allowed for efficient and accurate hypothesis testing and moderation effect analysis, which were crucial for understanding and

Table 4
Sample characteristics.

Factors		Frequency	%
Firm size	<50	41	13.6
	50–100	120	39.7
	100–300	141	46.7
Firm age	<5 years	50	16.6
	5–10	135	44.7
	≥11	117	38.7
Position	Vice president or above	167	55.3
	Director	89	29.5
	Manager	46	15.2
Industry	Online retailers in TikTok	302	100
Categories	Cosmetics and Personal Care	63	20.9
	Apparel	49	16.2
	Toys and Games	30	9.9
	Sports and Outdoor Activities	29	9.6
	Household Essentials	20	6.6
	Kitchenware	19	6.3
	Food and Beverages	35	11.6
	Mobile Phones and Digital Products	34	11.3
	Computers and Office Supplies	23	7.6

revealing the relationships among the variables, especially the moderating role of digital employee ethical anxiety (Singh et al., 2024).

While PLS-SEM excels at testing linear relationships and examining the net effects of variables, fsQCA offers unique insights into complex, nonlinear relationships and configurational patterns (Cheng et al., 2023; Ragin and Fiss, 2008). The combination of these methods provides a more comprehensive understanding of complex phenomena, as demonstrated in recent studies (Kim et al., 2024; Elshaer et al., 2024; Zheng and Li, 2024). This methodological triangulation enhances the robustness of findings by capturing both variance-based relationships and configurational patterns (Alnoor et al., 2024; Wang and Esperança, 2023). We employed Importance-Performance Map Analysis (IPMA) using SmartPLS 4 to provide deeper insights and recommendations. IPMA visually represents the importance and performance of various variables, helping to understand their impact on sustainable innovation performance. This analysis is critical for identifying key areas for managerial focus and improvement (Rauter et al., 2023).

FsQCA 4 software was used for sufficient condition analysis, revealing the complexity of causal relationships. fsQCA is a novel and powerful analytical method capable of uncovering configurations where multiple conditions must be simultaneously met to achieve a specific outcome (Kim et al., 2024). This method is particularly significant for exploring and understanding the configurational effects in digital transformation and sustainability contexts (Asiaei et al., 2023).

4.5. Common method bias

To mitigate common method bias, we employed both ex ante and ex post approaches (Podsakoff et al., 2003). Procedurally, we utilized different response formats, assured respondent anonymity, and counterbalanced question order. We also conducted Harman’s single-factor test, which revealed that the first factor accounted for 34.808% of the total variance, below the 50% threshold (Fuller et al., 2016). We applied the marker variable technique using “perceived ease of use” as a theoretically unrelated construct. The marker variable showed non-significant correlations with the study variables ($p > 0.05$), further suggesting common method bias was not a substantial concern.

4.6. Non-response bias

Non-response bias was examined by comparing early and late respondents on key study variables using independent samples t-tests (Wang and Esperança, 2023). No significant differences were found between early and late respondents in terms of firm size, firm age, industry, or respondent position. Chi-square tests also revealed no significant differences in the distribution of these variables between early and late respondents ($p > 0.05$). These results indicate that non-response bias is unlikely to be a major issue in our study.

4.7. Multicollinearity analysis

Multicollinearity was assessed by examining variance inflation factors (VIF). All VIF values were below 3, with a maximum of 2.564 and an average of 1.975, well below the recommended threshold of 5 (Hair et al., 2017). This suggests that multicollinearity is not a significant concern in our model.

5. Analysis and results

5.1. Measurement model

The measurement model was assessed through a comprehensive evaluation of reliability, convergent validity, and discriminant validity. As shown in Table 5, all constructs demonstrated satisfactory internal consistency reliability, with Cronbach’s alpha and composite reliability values exceeding the recommended threshold of 0.7 (Hair et al., 2017).

Table 5
Reliability and convergent validity.

Construct	Items	Loadings >0.7	VIF <3	Cronbach's Alpha >0.7	rho_a >0.7	rho_c >0.7	AVE >0.5	Mean	SD
Digital Employee Adoption (DEA)	DEA1	0.819	2.022	0.874	0.874	0.908	0.665	4.668	1.487
	DEA2	0.810	1.984						
	DEA3	0.826	2.113						
	DEA4	0.815	1.977						
	DEA5	0.808	1.914						
Exploitative Green Innovation (IGI)	IGI1	0.841	1.989	0.856	0.859	0.902	0.697	3.876	1.55
	IGI2	0.839	1.956						
	IGI3	0.815	1.909						
	IGI4	0.845	1.997						
Exploratory Green Innovation (RGI)	RGI1	0.830	1.893	0.856	0.857	0.903	0.698	4.461	1.533
	RGI2	0.849	2.041						
	RGI3	0.830	1.969						
	RGI4	0.834	1.947						
Future Sustainable Innovation Performance (FSIP)	FSIP1	0.809	2.193	0.919	0.922	0.934	0.639	4.445	1.503
	FSIP2	0.823	2.369						
	FSIP3	0.807	2.267						
	FSIP4	0.792	2.185						
	FSIP5	0.757	1.926						
	FSIP6	0.783	2.093						
	FSIP7	0.821	2.327						
	FSIP8	0.804	2.280						
Digital Employee Ethical Anxiety (DEEA)	DEEA1	0.839	2.564	0.934	0.945	0.945	0.681	4.353	1.497
	DEEA2	0.815	2.543						
	DEEA3	0.820	2.451						
	DEEA4	0.839	2.424						
	DEEA5	0.838	2.519						
	DEEA6	0.829	2.563						
	DEEA7	0.814	2.302						
	DEEA8	0.806	2.496						

The rho_a values, which provide a more accurate reliability estimate for PLS path modeling, also surpassed 0.7 for all constructs. Convergent validity was established as all item loadings were above 0.7, and the average variance extracted (AVE) for each construct exceeded 0.5.

Discriminant validity was assessed using three criteria: Fornell-Larcker criterion, heterotrait-monotrait (HTMT) ratio, and cross-loadings (Table 6) (Hair et al., 2017). The square root of AVE for each construct was greater than its correlations with other constructs, satisfying the Fornell-Larcker criterion. All HTMT ratios were below the conservative threshold of 0.85, further supporting discriminant validity. Each item's loading on its intended construct was higher than its cross-loadings on other constructs. Collectively, these results provide strong evidence for the reliability and validity of the measurement model, establishing a solid foundation for subsequent structural model analysis and hypothesis testing.

5.2. Structural model and hypothesis test

The structural model assessment was conducted using PLS-SEM (Hair et al., 2017). As illustrated in Fig. 3, the model demonstrates the hypothesized relationships among digital employee adoption, exploitative green innovation, exploratory green innovation, future sustainable innovation performance, and digital employee ethical anxiety. The path coefficients, standard deviations, t-values, p-values, and effect sizes (f^2) for each hypothesis are summarized in Table 7.

Digital employee adoption significantly influences both exploitative green innovation and exploratory green innovation, with path coefficients of 0.468 and 0.451, respectively, both significant at the $p < 0.001$ level. These results support hypotheses H1 and H2, indicating that digital employee adoption positively drives both types of green innovation within firms. Digital employee adoption also positively impacts future sustainable innovation performance, with a path coefficient of 0.238 ($p < 0.001$), supporting H3. Exploratory and exploitative green innovations significantly enhance future sustainable innovation performance, with path coefficients of 0.353 and 0.261, respectively, both significant at the $p < 0.001$ level. These findings support hypotheses H4

and H5, demonstrating that both types of green innovations are critical drivers of sustainable innovation performance. The effect sizes for these paths are 0.111 for exploratory green innovation to future sustainable innovation performance and 0.059 for exploitative green innovation to future sustainable innovation performance, highlighting their importance in achieving sustainable outcomes.

5.3. Moderating effect

Moderating effect analysis was performed using the bootstrapping algorithm (Hair et al., 2017). Digital employee ethical anxiety negatively moderates the relationship between digital employee adoption and both types of green innovation (Figs. 4, 5, and 6). The interaction effect for digital employee adoption and exploitative green innovation is -0.446 ($p < 0.001$), while for digital employee adoption and exploratory green innovation, it is -0.453 ($p < 0.001$), supporting hypotheses H6 and H8. These negative moderations indicate that higher levels of digital employee ethical anxiety weaken the positive impact of digital employee adoption on green innovation. The effect sizes are 0.053 for the moderation on exploitative green innovation and 0.018 on exploratory green innovation, suggesting a notable but varying degree of moderation.

Interestingly, digital employee ethical anxiety positively moderates the relationship between digital employee adoption and future sustainable innovation performance with a path coefficient of 0.143 ($p < 0.05$) (Fig. 5), contrary to H7. This indicates that while digital employee ethical anxiety poses challenges, it can also drive firms to adopt more sustainable practices, enhancing overall performance.

However, digital employee ethical anxiety does not significantly moderate the relationships between exploitative and exploratory green innovations and future sustainable innovation performance. The interaction effects of digital employee ethical anxiety with exploitative green innovation and exploratory green innovation on future sustainable innovation performance were 0.035 ($p = 0.582$) and 0.010 ($p = 0.887$), respectively, providing no support for hypotheses H9 and H10.

Table 6

Fornell-Larcker Criterion, Heterotrait-Monotrait Ratio and Cross loadings.

Construct	DEA	DEEA	FSIP	IGI	RGI
DEA	0.815	0.149	0.583	0.568	0.550
DEEA	0.133	0.825	0.109	0.073	0.047
FSIP	0.525	0.096	0.800	0.577	0.624
IGI	0.492	−0.020	0.518	0.835	0.715
RGI	0.476	−0.023	0.558	0.613	0.836
Items	DEA	DEEA	FSIP	IGI	RGI
DEA1	0.819	0.155	0.413	0.390	0.381
DEA2	0.810	0.062	0.443	0.391	0.362
DEA3	0.826	0.046	0.421	0.384	0.412
DEA4	0.815	0.132	0.410	0.446	0.379
DEA5	0.808	0.145	0.453	0.393	0.405
DEEA1	0.148	0.839	0.114	0.017	0.036
DEEA2	0.085	0.815	0.093	−0.007	−0.044
DEEA3	0.059	0.820	0.057	−0.045	−0.035
DEEA4	0.099	0.839	0.115	−0.047	−0.016
DEEA5	0.080	0.838	0.041	−0.012	−0.021
DEEA6	0.124	0.829	0.097	−0.008	−0.005
DEEA7	0.156	0.814	0.060	0.005	−0.030
DEEA8	0.119	0.806	0.043	−0.040	−0.060
FSIP1	0.443	0.026	0.809	0.491	0.513
FSIP2	0.397	0.039	0.823	0.445	0.490
FSIP3	0.389	0.082	0.807	0.421	0.463
FSIP4	0.445	0.103	0.792	0.375	0.425
FSIP5	0.392	0.142	0.757	0.357	0.367
FSIP6	0.385	0.046	0.783	0.408	0.416
FSIP7	0.514	0.093	0.821	0.426	0.463
FSIP8	0.382	0.099	0.804	0.375	0.412
IGI1	0.446	0.043	0.424	0.841	0.542
IGI2	0.419	−0.058	0.445	0.839	0.492
IGI3	0.379	−0.075	0.357	0.815	0.497
IGI4	0.397	0.015	0.493	0.845	0.516
RGI1	0.396	−0.011	0.477	0.505	0.830
RGI2	0.396	−0.009	0.504	0.537	0.849
RGI3	0.384	−0.022	0.437	0.480	0.830
RGI4	0.413	−0.035	0.443	0.524	0.834

Note: Bold numbers show construct element loading values.

Note: Bold numerals are AVE square roots. Below the diagonal in the lower-left corner are Fornell-Larcker Criterion figures, while matching spots show HTMT Ratio analysis results.

5.4. PLS-SEM model fit, R^2 , Q^2 , f^2 and CVPAT

The assessment of the PLS-SEM model fit, along with R^2 , Q^2 , and f^2 values, provides comprehensive insights into the robustness and explanatory power of the proposed model. The model fit indices indicate an excellent fit, with an SRMR (Standardized Root Mean Square Residual) value of 0.043, which is well below the acceptable threshold of 0.08 (Hair et al., 2017). The d_ULS and d_G values are 1.217 and 0.472, respectively, further affirming the model's adequacy. The Chi-square

value of 826.882 and an NFI of 0.867 also suggest a good fit, indicating that the model is well-specified and captures the underlying data structure effectively.

The R^2 values provide insight into the variance explained by the endogenous constructs. The R^2 value for future sustainable innovation performance is 0.457, indicating that the model explains 45.7% of the variance in future sustainable innovation performance. This substantial level of explained variance underscores the model's predictive relevance. The R^2 values for exploitative green innovation and exploratory green innovation are 0.443 and 0.432, respectively, demonstrating the significant impact of digital employee adoption on these innovation outcomes. In contrast, digital employee ethical anxiety shows a minimal R^2 value of 0.011, suggesting a limited direct influence within the model's scope.

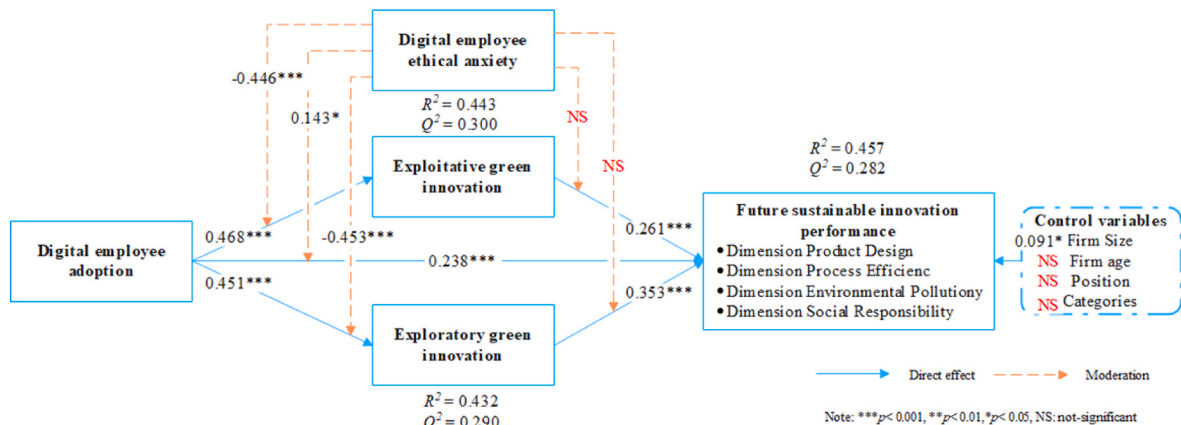
The Q^2 values, which measure the model's predictive accuracy through cross-validated redundancy, further validate the model's robustness. The Q^2 values for future sustainable innovation performance, exploitative green innovation, and exploratory green innovation are 0.282, 0.300, and 0.290, respectively, indicating that the model has substantial predictive relevance for these constructs. The digital employee ethical anxiety construct, however, exhibits a Q^2 value of 0.006, signifying negligible predictive accuracy.

Effect size (f^2) analysis provides additional insights into the relative importance of each construct within the model. The f^2 values indicate that digital employee adoption has a moderate to large effect on both exploitative green innovation (0.383) and exploratory green innovation (0.350), as well as a small effect on future sustainable innovation performance (0.053). The digital employee ethical anxiety construct exhibits negligible effect sizes across all dependent variables, with values

Table 7

Hypothesis test.

Hypotheses	β	SD	T	p	f^2	Support?
H1: DEA → IGI	0.468	0.045	10.283	0.000	0.383	✓
H2: DEA → RGI	0.451	0.045	9.976	0.000	0.350	✓
H3: DEA → FSIP	0.238	0.061	3.900	0.000	0.053	✓
H4: RGI → FSIP	0.353	0.062	5.662	0.000	0.111	✓
H5: IGI → FSIP	0.261	0.062	4.202	0.000	0.059	✓
H6: DEEA × DEA → IGI	−0.446	0.049	9.078	0.000	0.053	✓
H7: DEEA × DEA → FSIP	0.143	0.063	2.263	0.024	0.013	× *
H8: DEEA × DEA → RGI	−0.453	0.050	9.065	0.000	0.018	✓
H9: DEEA × IGI → FSIP	0.035	0.064	0.550	0.582	0.000	×
H10: DEEA × RGI → FSIP	0.010	0.069	0.141	0.887	0.000	×

**Fig. 3.** PLS-SEM model.

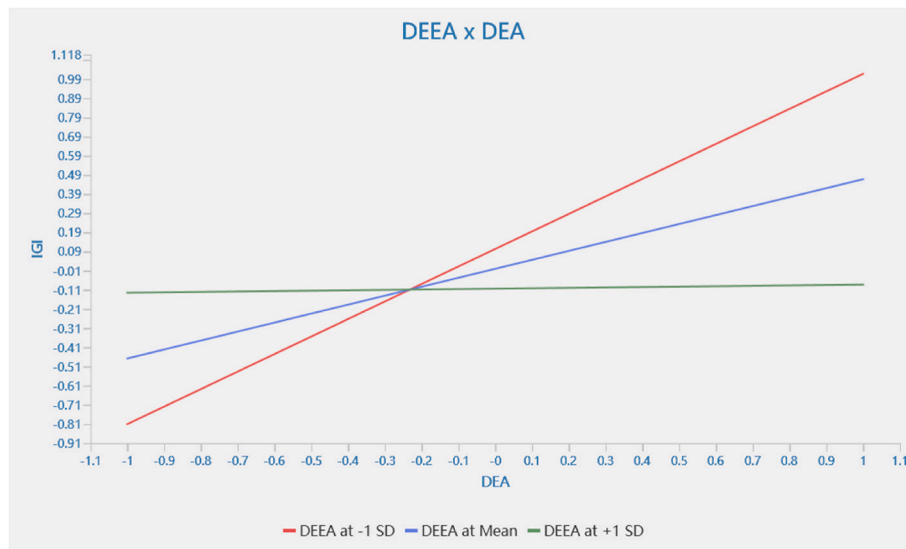


Fig. 4. Moderating effect of digital employee ethical anxiety on digital employee adoption and exploitative green innovation. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

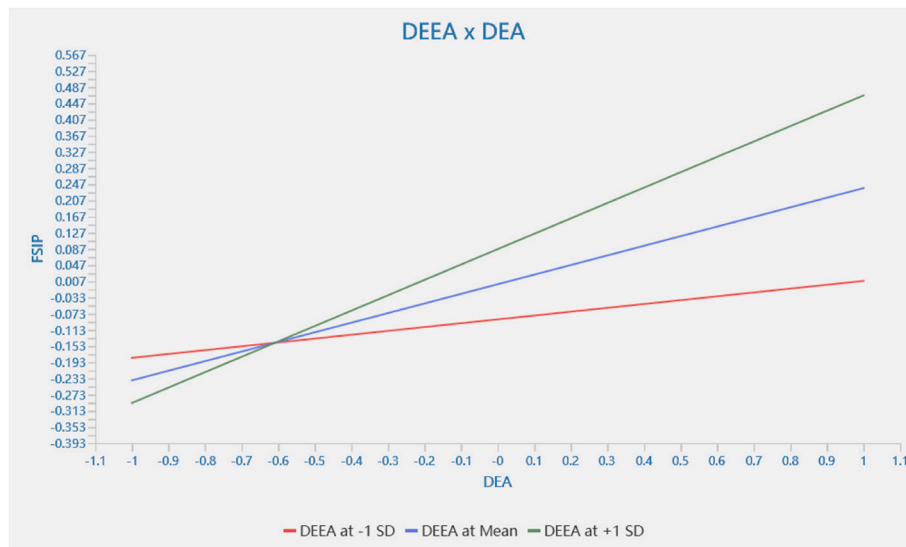


Fig. 5. Moderating effect of digital employee ethical anxiety on digital employee adoption and future sustainable innovation performance.

ranging from 0.013 to 0.019, highlighting its limited moderating influence. The exploitative green innovation and exploratory green innovation constructs show small to moderate effect sizes on future sustainable innovation performance, with values of 0.059 and 0.111, respectively, reinforcing their crucial roles in driving sustainable innovation performance.

The CVPAT results (Table 8) demonstrate that our PLS-SEM model outperforms both the indicator average benchmark and linear model benchmark, with an overall significant improvement in predictive power ($t = 3.675$, $p < 0.001$) (Sharma et al., 2022). This additional analysis strengthens our confidence in the model's predictive capabilities, particularly for constructs with homogeneous factor loadings (Liengaard et al., 2021).

Our PLS-SEM model generally performs better or comparably to the IA method and significantly better than the LM in predicting the constructs (Capeau et al., 2024). The low average loss differences and associated t-values suggest that our model has adequate predictive power, especially considering items with homogenous factor loadings.

5.5. CB-SEM dual test

To validate the robustness of our PLS-SEM results, we conducted a covariance-based structural equation modeling (CB-SEM) analysis as a complementary approach (Hair et al., 2017; Wang and Zhang, 2025). The CB-SEM analysis was performed using SmartPLS 4.0 software with maximum likelihood estimation. As shown in Table 9, the model fit indices demonstrate excellent fit between the theoretical model and empirical data. Specifically, the chi-square/degrees of freedom ratio (1.454) is well below the recommended threshold of 3.0. The root mean square error of approximation (RMSEA = 0.039) indicates close fit, while both the goodness-of-fit index (GFI = 0.925) and adjusted goodness-of-fit index (AGFI = 0.906) exceed the 0.90 criterion for good fit. The standardized root mean square residual (SRMR = 0.07) falls below the conservative cutoff of 0.08, suggesting minimal discrepancies between observed and model-implied covariances. The normed fit index (NFI = 0.928), Tucker-Lewis index (TLI = 0.973), and comparative fit index (CFI = 0.976) all surpass the 0.90 benchmark, providing strong evidence of good model fit. Collectively, these results corroborate the

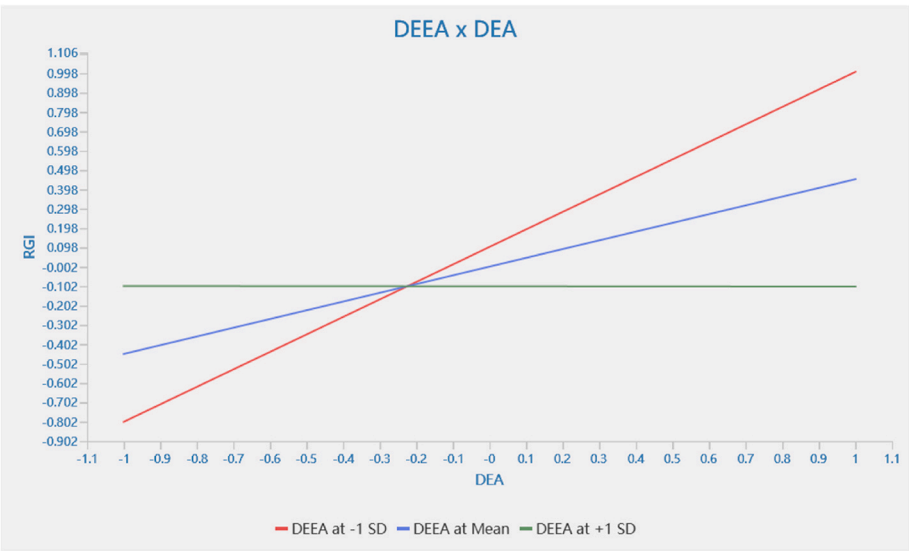


Fig. 6. Moderating effect of digital employee ethical anxiety on digital employee adoption and exploratory green innovation. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 8
CVPAT results.

Construct	CVPAT- PLS-SEM-Vs Indicator average					CVPAT- PLS-SEM-vs linear model				
	PLS loss	IA loss	Average loss difference	t value	p value	PLS loss	LM loss	Average loss difference	t value	p value
DEA	3.377	3.363	0.014	1.684	0.093	3.377	3.484	−0.107	2.823	0.005
DEEA	3.320	3.326	−0.007	0.255	0.799	3.320	3.357	−0.037	0.699	0.485
FSIP	3.552	3.560	−0.008	0.196	0.845	3.552	3.636	−0.084	3.971	0.000
IGI	3.479	3.465	0.013	1.451	0.148	3.479	3.581	−0.102	2.778	0.006
RGI	3.414	3.401	0.013	2.511	0.013	3.414	3.489	−0.075	1.523	0.129
Overall	3.429	3.427	0.002	0.121	0.904	3.429	3.505	−0.076	3.675	0.000

The CvPAT results show that: The model demonstrates superior predictive ability compared to linear models for Digital Employee Adoption ($t = 2.823$, $p = 0.005$); Future Sustainable Innovation Performance predictions are significantly better than linear models ($t = 3.971$, $p < 0.001$); Exploitative Green Innovation shows improved predictive power ($t = 2.778$, $p = 0.006$).

Table 9
CB-SEM dual test.

Index	Estimated Model	Reference	Result
ChiSqr/df	1.454	<3	Good
RMSEA	0.039	<0.08	Good
GFI	0.925	>0.90	Good
AGFI	0.906	>0.90	Good
PGFI	0.737	>0.50	Good
SRMR	0.070	<0.08	Good
NFI	0.928	>0.90	Good
TLI	0.973	>0.90	Good
CFI	0.976	>0.90	Good

validity of our measurement model and lend further credence to the structural relationships identified through PLS-SEM analysis.

5.6. Importance-performance map analysis

The importance-performance map analysis (IPMA) provides critical insights into the relative impact and performance of key constructs in the model (Fig. 7) (Hair et al., 2017). Digital employee adoption emerges as the most influential factor, with the highest importance (0.527) and performance (61.572) scores. This underscores digital employee adoption’s pivotal role in driving sustainable innovation outcomes. Exploratory green innovation demonstrates the second-highest importance (0.371) and above-average performance (57.861), highlighting its significant contribution. Exploitative green

innovation shows moderate importance (0.251) but lower performance (48.028), suggesting potential for improvement. Notably, digital employee ethical anxiety exhibits minimal importance (0.030) despite above-average performance (56.755), indicating a limited impact on sustainable innovation performance in this context. These findings offer nuanced guidance for managerial focus, emphasizing the need to prioritize digital employee adoption and exploratory innovation efforts while addressing performance gaps in exploitative innovation practices.

5.7. fsQCA

To complement the PLS-SEM findings, this study employed fuzzy-set Qualitative Comparative Analysis (fsQCA) to explore the complex causal relationships between digital employee adoption, ambidextrous green innovation, ethical anxiety, and sustainable innovation performance (Kim et al., 2024; Wang and Zhang, 2024). While PLS-SEM assumes linear and symmetrical relationships, fsQCA allows for the examination of asymmetric, equifinal, and conjunctural causation (Ragin and Fiss, 2008).

The first step in fsQCA involved calibrating the causal conditions and outcome. The 7-point Likert scale values were transformed into fuzzy membership scores ranging from 0.05 (full non-membership) to 0.95 (full membership), with 0.5 as the crossover point (Ragin, 2008). Necessity analysis revealed no single construct with a consistency value exceeding 0.9, indicating the absence of necessary conditions (Ragin and Fiss, 2008).

A truth table was constructed to display all possible causal

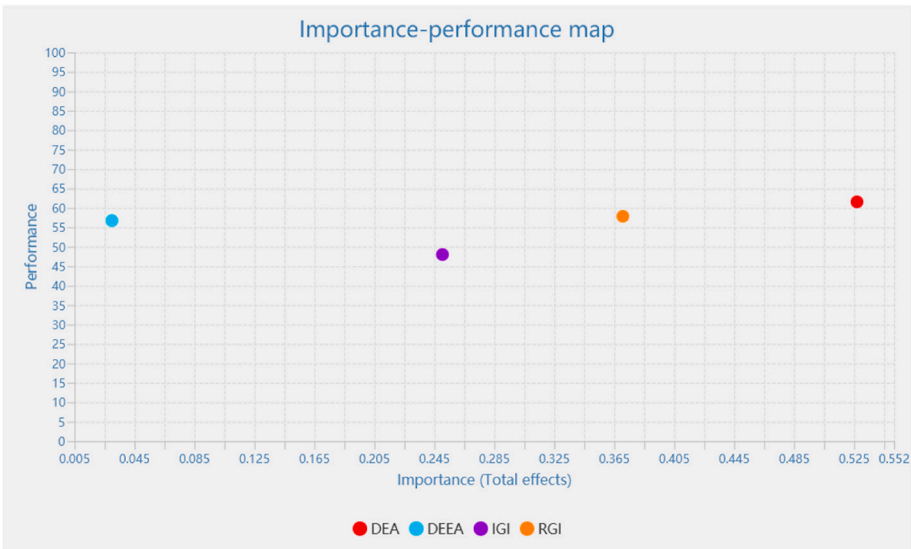


Fig. 7. Ipma.

combinations. Following established guidelines (Ragin and Fiss, 2008), configurations with a minimum frequency of 3 cases and a raw consistency threshold of 0.80 were retained for further analysis. The fsQCA analysis produced three solutions: complex, parsimonious, and intermediate. The intermediate solution was selected for interpretation due to its balance between complexity and parsimony. Table 10 presents three configurations leading to high levels of future sustainable innovation performance. The overall solution consistency (0.859) and coverage (0.611) indicate that the identified configurations are empirically relevant and consistent in producing the outcome (Ragin and Fiss, 2008). Notably, exploratory green innovation appears as a core condition across all three configurations, underscoring its critical role in driving sustainable innovation performance.

Configuration 1 combines high levels of digital employee ethical anxiety with both exploitative and exploratory green innovation, suggesting that ethical concerns may stimulate innovative practices when coupled with ambidextrous innovation strategies. Configuration 2 highlights the synergistic effect of digital employee adoption and ambidextrous green innovation in the absence of high ethical anxiety. Configuration 3 reveals an intriguing combination of high digital employee adoption, high ethical anxiety, and high exploratory green innovation, indicating that ethical concerns may enhance the relationship between digital adoption and exploratory innovation under certain conditions. These findings extend our understanding of the complex interplay between digital employee adoption, ethical considerations, and sustainable innovation performance, offering nuanced insights beyond the linear relationships identified in the PLS-SEM analysis.

Table 10
fsQCA.

Variable	Configurations		
	1	2	3
DEA		●	●
DEEA	●		●
IGI	●	●	
RGI	●	●	●
Raw coverage	0.466	0.555	0.457
Unique coverage	0.032	0.121	0.024
Consistency	0.900	0.870	0.887
Solution coverage	0.611		
Solution consistency	0.859		

Note: The '●' symbol indicates core conditions. Lack of symbols in a cell indicates 'irrelevant' or 'not important' context.

6. Practical examples of future sustainable innovation performance with digital employees

6.1. Research methodology for practical examples

This study employed a qualitative research methodology to gather practical examples through in-depth interviews with senior executives from online retailers on TikTok (Behera et al., 2024). Following established qualitative research practices, we conducted semi-structured interviews with four senior executives from different retail categories between May and July 2024 (Jauhari et al., 2024). The interviewees were selected based on their extensive experience in implementing digital employees and sustainable innovation initiatives in their organizations (Nosi et al., 2023). The interviews were conducted online, lasting approximately 61–89 min each. All interviews were recorded with permission and transcribed for analysis (Xu et al., 2024). To ensure reliability and validity, we followed a systematic interview protocol focusing on three key areas: (1) the implementation of digital employees, (2) sustainable innovation practices, and (3) observed outcomes and challenges.

The selection of interviewees was based on purposive sampling to ensure representation across different retail categories and firm sizes (Bhardwaj et al., 2023; Lim et al., 2022). We specifically targeted firms that had successfully implemented digital employee initiatives for at least one year and demonstrated commitment to sustainable innovation. Table 11 presents the characteristics of the interviewees, including their industry categories, work experience, positions, and specific application areas of digital employees in their organizations.

6.2. Enhancing customer engagement with digital employees

Interview1 is a prominent online retailer in the Cosmetics and Personal Care category. With over ten years of experience in the industry, the firm has actively adopted digital employees to enhance customer engagement and streamline operations. The CEO highlighted that digital employees have significantly improved the firm's ability to interact with customers through personalized product recommendations and real-time customer support. This strategic move has not only increased customer satisfaction but also reduced operational costs. The integration of digital employees has enabled the firm to implement more efficient and sustainable business practices, such as reducing paper waste through digital receipts and optimizing inventory management to minimize excess stock and associated waste. These efforts have

Table 11
Characteristics of the interviewees.

ID	Categories	Work years	Position	Application areas	Advantages
Interview1	Cosmetics and Personal Care	10+	CEO	Customer engagement, inventory management	Increased customer satisfaction, reduced operational costs, optimized inventory, reduced paper waste
Interview2	Apparel	8	Vice President of Operations	Supply chain forecasting, logistics optimization	Reduced resource wastage, minimized carbon emissions, improved supplier coordination, enhanced market competitiveness
Interview3	Toys and Games	6	Director of Product Innovation	Accelerated design and prototyping, eco-friendly materials	Rapid product development, efficient experimentation with sustainable materials, market leadership in sustainable products
Interview4	Food and Beverages	15+	Manager of Sustainability Initiatives	Energy-efficient production, waste reduction programs	Focus on strategic sustainability initiatives, significant environmental impact, enhanced reputation among environmentally conscious consumers

culminated in a notable improvement in the firm's sustainable innovation performance.

6.3. Streamlining supply chain operations

Interview2 operates in the Apparel category and has been in business for eight years. The Vice President of Operations emphasized the role of digital employees in streamlining supply chain operations. By leveraging AI-powered virtual assistants, the firm has enhanced its ability to forecast demand, manage inventory, and optimize logistics. This integration has led to a significant reduction in resource wastage and carbon emissions associated with transportation. The firm's digital employees have facilitated better coordination with suppliers, ensuring that sustainable practices are upheld throughout the supply chain. The Vice President noted that these improvements have not only bolstered the firm's sustainability credentials but also enhanced its competitive edge in the market.

6.4. Innovating product development processes

Interview3 is a leading player in the Toys and Games industry, with a firm age of 6 years. The Director of Product Innovation highlighted the transformative impact of digital employees on the firm's product development processes. Digital employees have been instrumental in accelerating the design and prototyping phases by providing rapid simulations and feedback. This capability has allowed the firm to experiment with eco-friendly materials and designs more efficiently. The Director mentioned that the use of digital employees has led to the development of several new sustainable products that have been well-received in the market. These innovations have significantly contributed to the firm's overall sustainable innovation performance, positioning it as a forward-thinking leader in the industry.

6.5. Fostering a culture of sustainability

Interview4 is a well-established firm in the Food and Beverages category, with over 15 years of experience. The firm's Manager of Sustainability Initiatives discussed how digital employees have fostered a culture of sustainability within the organization. By automating routine tasks, digital employees have freed up human workers to focus on strategic sustainability initiatives. The Manager noted that this shift has led to the implementation of various green practices, such as energy-efficient production processes and waste reduction programs. The firm's commitment to sustainability, supported by the capabilities of digital employees, has resulted in significant improvements in its sustainable innovation performance. This holistic approach has not only enhanced the firm's environmental impact but also its reputation among environmentally conscious consumers.

7. Discussion

7.1. Interpretation of findings

Our study addressed three key research questions examining the relationships between digital employee adoption, ambidextrous green innovation, and future sustainable innovation performance in retail and consumer services. Regarding our first research question about how digital employee adoption influences exploitative and exploratory green innovation, our findings demonstrate significant positive effects ($\beta = 0.468$, $p < 0.001$ for exploitative; $\beta = 0.451$, $p < 0.001$ for exploratory green innovation), confirming that digital employee adoption enhances both types of innovation in retail settings. For our second research question concerning the mediating role of ambidextrous green innovation, the results confirm that both exploitative ($\beta = 0.261$, $p < 0.001$) and exploratory ($\beta = 0.353$, $p < 0.001$) green innovation significantly mediate the relationship between adoption and future sustainable innovation performance. Finally, addressing our third research question about the moderating role of digital employee ethical anxiety, we found that it negatively moderates the relationships between adoption and both types of green innovation, while unexpectedly strengthening the direct link between adoption and future sustainable innovation performance. These findings collectively achieve our research objective of elucidating the complex mechanisms through which digital employees influence sustainable innovation in retail and consumer services.

The empirical results of this study provide strong support for the proposed Adoption-Innovations-Future Performance (AIF) model, offering nuanced insights into the complex relationships between digital employee adoption, ambidextrous green innovation, and future sustainable innovation performance. First, our finding that digital employee adoption positively influences both exploitative and exploratory green innovation ($\beta = 0.468$ and $\beta = 0.451$ respectively, $p < 0.001$) aligns with Singh et al. (2024) and Ardito (2023), who similarly found that digital technologies enhance organizational innovation capabilities. However, our study extends their work by specifically demonstrating these effects in the context of digital employees in retail settings. The findings reveal that digital employee adoption significantly enhances both exploitative and exploratory green innovation, as well as directly improving future sustainable innovation performance. This aligns with prior research suggesting that digital technologies can serve as catalysts for sustainable innovation by augmenting organizational capabilities and fostering novel approaches to environmental challenges (Mehmood et al., 2024a,b).

Second, our results showing the mediating role of ambidextrous green innovation support and extend Wang et al.'s (2023) findings on how digital platform capabilities improve sustainable innovation performance. While they focused on open innovation as a mediator, we demonstrate that both exploitative and exploratory green innovations serve as crucial mediating mechanisms. The study finds that ambidextrous green innovation, comprising both exploitative and exploratory dimensions, plays a crucial mediating role in translating digital

employee adoption into enhanced future sustainable innovation performance. This supports the notion that firms must balance incremental improvements with radical innovations to achieve long-term sustainability goals (Sun et al., 2024). The stronger effect of exploratory green innovation on future sustainable innovation performance, compared to exploitative innovation, underscores the importance of pursuing breakthrough eco-friendly technologies and entering new green markets for sustained competitive advantage (Cheng et al., 2023).

Third, our unexpected finding that digital employee ethical anxiety positively moderates the relationship between adoption and future sustainable innovation performance ($\beta = 0.143$, $p < 0.05$) presents an interesting contrast to Li and Huang's (2020) work, which primarily viewed ethical anxiety as a barrier. Our results suggest a more nuanced role where ethical concerns can actually enhance sustainable outcomes. The moderating effect of digital employee ethical anxiety reveals a more nuanced picture than initially hypothesized. While ethical anxiety negatively moderates the relationship between digital employee adoption and both types of green innovation, it unexpectedly strengthens the direct positive link between adoption and future sustainable innovation performance. This paradoxical finding suggests that ethical concerns may stimulate more thoughtful and responsible innovation practices, ultimately leading to improved sustainability outcomes (Singh et al., 2024). It aligns with recent research highlighting the potential of ethical considerations to drive more robust and truly sustainable innovations in the long run (Xie et al., 2024).

Fourth, our fsQCA results reveal three distinct configurations leading to high sustainable innovation performance that align with Cheng et al.'s (2023) configurational findings, though we specifically identify new patterns in the retail context. The consistent presence of exploratory green innovation across all configurations reinforces its critical role in driving sustainable outcomes. Moreover, the synergistic effects observed between digital employee adoption, ethical anxiety, and innovation strategies in certain configurations highlight the complex, non-linear nature of sustainable innovation processes (Rauter et al., 2023).

These findings extend previous research on digital transformation and sustainable innovation by elucidating the specific mechanisms through which digital employees influence organizational outcomes. They also contribute to the growing body of literature on the ethical implications of AI adoption in the workplace, offering a more nuanced perspective on how ethical concerns can both hinder and enhance innovation processes (Al Halbusi et al., 2024). The results underscore the need for a holistic approach to digital employee integration, one that balances technological capabilities with ethical considerations and strategic innovation management.

7.2. Theoretical implications

This study makes several theoretical contributions to the literature on digital transformation, sustainable innovation, and AI ethics, particularly in the context of retail and consumer services.

First, we introduce the novel Adoption-Innovations-Future Performance (AIF) model, which extends the traditional Antecedents-Behaviour-Consequences (ABC) framework in a unique way tailored to digital transformation. The AIF model elucidates the complex relationships between digital employee adoption, ambidextrous green innovation (both exploitative and exploratory), and future sustainable innovation performance. By integrating digital employees as key antecedents and highlighting the mediating roles of exploitative (H1, H5) and exploratory green innovation (H2, H4), the model provides a comprehensive lens through which to understand how digital technologies drive sustainable organizational outcomes. This theoretical advancement fills a crucial gap in the literature by offering a nuanced understanding of the mechanisms underpinning digital transformation in retail settings (Singh et al., 2024; Ardito, 2023).

Second, we identify a new phenomenon regarding the paradoxical role of digital employee ethical anxiety. While previous research

predominantly viewed ethical anxiety as a barrier to innovation, our findings reveal its dual nature—acting both as an inhibitor of green innovation and an enhancer of sustainable performance. Specifically, ethical anxiety negatively moderates the relationships between digital employee adoption and both exploitative (H6) and exploratory green innovation (H8), suggesting it can dampen innovation efforts due to concerns over ethical implications (Li and Huang, 2020; Gkinko and Elbanna, 2023). Conversely, it positively moderates the direct link between digital employee adoption and future sustainable innovation performance (H7), indicating that ethical concerns can motivate firms to adopt more responsible and sustainable practices, ultimately enhancing performance (Xie et al., 2024). This novel finding challenges existing theoretical assumptions about the role of ethical concerns in digital transformation and underscores the complexity of ethical anxiety's impact on organizational outcomes.

Third, our research introduces a new conceptualization of digital employees as quasi-team members rather than mere technological tools. This perspective significantly contributes to the theoretical understanding of human-AI collaboration in retail settings. By framing digital employees as integral collaborators who augment human capabilities, we provide fresh insights into their role in both enhancing operational efficiency and fostering innovation (Le et al., 2024). This conceptualization underpins our hypotheses regarding digital employee adoption's impact on exploitative and exploratory green innovation (H1-H3) and its interactions with ethical anxiety (H6-H8). It advances the discourse on AI integration by highlighting the symbiotic relationship between human employees and AI agents in driving sustainable innovation (Van Doorn et al., 2023).

Fourth, we develop a novel configurational approach to understanding sustainable innovation performance through our use of fuzzy-set Qualitative Comparative Analysis (fsQCA). This methodology reveals previously unidentified pathways to sustainability success by showing that multiple configurations of digital employee adoption, ambidextrous green innovation, and ethical anxiety can lead to high sustainable innovation performance (Cheng et al., 2023; Ragin and Fiss, 2008). Our fsQCA results align with the concept of equifinality in organizational studies, demonstrating that different combinations of conditions can achieve similar outcomes (H1-H10). This theoretical contribution emphasizes the importance of context-specific strategies and the complex interplay of factors in digital transformation initiatives, offering a more holistic understanding than traditional variance-based analyses.

Finally, by directly correlating our theoretical contributions with our hypotheses, we enhance the coherence and explanatory power of our study. Each novel insight is linked to specific hypotheses: the AIF model relates to H1-H5, the paradoxical role of ethical anxiety to H6-H8, the conceptualization of digital employees as quasi-team members to H1-H3 and H6-H8, and the configurational approach to H1-H10. This alignment reinforces the validity and robustness of our findings. Overall, our work not only expands theoretical discourse but also provides actionable insights for retailers aiming to leverage digital employees effectively for sustainable innovation, bridging the gap between theory and practice.

7.3. Practical implications for retailers and service providers

This study presents several practical implications for retailers and service providers aiming to leverage digital employees to enhance sustainable innovation performance. Our findings, directly linked to the hypotheses, offer actionable insights that can inform strategic decision-making in the industry.

First, consistent with Hypotheses H1 and H2, the significant positive relationships between digital employee adoption and both exploitative ($\beta = 0.468$, $p < 0.001$) and exploratory green innovation ($\beta = 0.451$, $p < 0.001$) underscore the necessity for retailers to strategically adopt and integrate digital employees into customer-facing operations. Retailers should perceive digital employees not merely as technological tools but

as quasi-team members who augment human capabilities in novel ways to enrich customer experiences (Kumar et al., 2024). By fostering this integration, firms can refine existing sustainable practices (exploitative innovation) while simultaneously exploring breakthrough eco-friendly technologies (exploratory innovation) that can transform the shopping experience (Asiaei et al., 2023). Practically, this entails investing in AI-powered virtual agents capable of analyzing customer data to optimize current green products and services and identifying opportunities for novel sustainable offerings.

Second, addressing Hypotheses H4 and H5, the mediating role of ambidextrous green innovation highlights the importance of balancing exploitative and exploratory innovation strategies to maximize sustainable innovation performance. Retailers must develop capabilities that enable both incremental improvements and radical innovations. This ambidextrous approach allows firms to respond effectively to immediate sustainability demands while positioning themselves for future market shifts. For practitioners, this involves allocating resources and cultivating organizational structures that support both types of innovation, such as dedicated teams for process optimization and experimental units for pioneering sustainable technologies.

Third, in line with Hypotheses H6 and H8, the negative moderating effect of digital employee ethical anxiety on the relationship between digital employee adoption and green innovation reveals that ethical concerns can dampen innovation efforts. However, importantly, as per the findings related to Hypothesis H7, ethical anxiety positively moderates the direct link between digital employee adoption and future sustainable innovation performance. This paradox suggests that while ethical anxiety may initially impede innovation processes, it can ultimately drive firms toward more thoughtful and responsible innovation practices that enhance long-term sustainability performance and consumer trust (Singh et al., 2024). Consequently, retail managers should not regard ethical anxiety merely as an obstacle but as a catalyst for developing robust and genuinely sustainable innovations in their service delivery. Practically, this involves proactively addressing ethical concerns through transparent communication with customers, comprehensive employee training on AI ethics, and establishing clear ethical guidelines for AI use in customer interactions (Zhou et al., 2024).

Fourth, the configurational analysis findings, consistent with the fsQCA results, provide valuable insights into strategic decision-making for retail managers (Cheng et al., 2023). The identification of multiple configurations leading to high sustainable innovation performance indicates that there is no one-size-fits-all approach. Managers should recognize that the optimal strategy depends on the specific retail context and resource configurations. By carefully considering the interplay between digital employee adoption, ethical considerations, and innovation strategies, retailers can develop tailored approaches that maximize their sustainable innovation performance while enhancing customer satisfaction. This might involve, for example, leveraging high levels of exploratory green innovation in conjunction with digital employee adoption, even in the presence of ethical anxiety, to achieve superior sustainability outcomes.

Lastly, the practical examples provided in this study illustrate the transformative potential of digital employees across various retail and service industries. By enhancing customer engagement, streamlining supply chain operations, accelerating product development, and fostering a culture of sustainability, digital employees can be leveraged to address environmental challenges and create customer value in novel ways (Lee and Roh, 2023). Retailers across sectors are encouraged to explore opportunities to integrate digital employees into their sustainability initiatives. This integration not only meets regulatory and societal expectations but also achieves strategic advantages in an increasingly eco-conscious market. For instance, firms can utilize digital employees to implement energy-efficient production processes, reduce waste, and develop innovative sustainable products that resonate with environmentally conscious consumers.

In conclusion, our findings offer a comprehensive understanding of

how digital employee adoption, mediated by ambidextrous green innovation and moderated by ethical anxiety, impacts future sustainable innovation performance. Retailers and service providers can harness these insights to develop and implement effective strategies that align technological advancements with sustainability objectives. By doing so, they can enhance their performance, fulfill corporate social responsibilities, and gain a competitive edge in the market.

7.4. Policy implications for retail and service industries

The findings of this study offer several important policy implications for governments and regulatory bodies seeking to promote sustainable innovation through digital transformation in retail and service industries. Policymakers should prioritize the development of comprehensive frameworks that encourage the responsible adoption of digital employees in customer-facing roles while addressing associated ethical concerns. Such frameworks could include guidelines for transparency in AI decision-making processes in retail settings and safeguards against potential job displacement in customer service roles (Xie et al., 2024). Policies should be designed to incentivize retailers' investments in both exploitative and exploratory green innovations, recognizing their complementary roles in driving sustainable performance and enhancing customer experiences (Asiaei et al., 2023).

Given the moderating effect of digital employee ethical anxiety, policymakers should consider mandating ethics training programs for retail organizations implementing AI technologies in customer service. These programs could help mitigate anxiety while fostering a culture of responsible innovation in retail settings (Li and Huang, 2020). Regulatory bodies should work towards establishing industry-wide standards for the ethical development and deployment of digital employees in retail, ensuring a level playing field for firms and building public trust in these technologies used in customer interactions (Singh et al., 2024).

To support the transition towards sustainable digital innovation in retail and services, governments could introduce targeted funding schemes and tax incentives for firms demonstrating ambidextrous green innovation capabilities in their customer offerings. Such policies could be particularly beneficial for SME retailers, which often face resource constraints in pursuing digital transformation and sustainability initiatives simultaneously (Taghizadeh et al., 2023). Moreover, policymakers should consider creating platforms for knowledge sharing and collaboration between academia, retail industry, and government to accelerate the development and diffusion of sustainable digital innovations in customer service (Wang et al., 2023).

Lastly, given the global nature of sustainability challenges and digital transformation in retail, international cooperation in policy development is crucial. Harmonizing regulations and standards across jurisdictions could facilitate the scaling of sustainable digital innovations in retail and prevent regulatory arbitrage in customer service practices (Ardito, 2023). By implementing these policy measures, governments can create an enabling environment for retailers to leverage digital employees in pursuit of sustainable innovation, ultimately contributing to improved customer experiences and broader societal and environmental goals.

7.5. Limitations, and future scope

While this study makes significant contributions to understanding digital employee adoption and sustainable innovation in retail settings, several limitations warrant acknowledgment and provide opportunities for future research. First, the cross-sectional nature of our data collection presents a methodological limitation that constrains causal inferences. Although our two-wave survey design helped mitigate some temporal concerns, future studies should employ longitudinal approaches to better capture the dynamic evolution of digital employee adoption and its impact on sustainable innovation performance over time. Specifically, researchers could conduct multi-year panel studies tracking

retailers' digital transformation journeys, measuring changes in innovation outcomes at regular intervals to establish stronger causal relationships.

Second, our sample's focus on TikTok retailers in China may limit the generalizability of findings to other retail contexts and cultural settings. Future research should validate our findings across different retail platforms (e.g., Amazon, Instagram) and geographical regions to enhance external validity. Cross-cultural comparative studies could illuminate how different institutional environments and cultural values influence the relationship between digital employee adoption and sustainable innovation. Such research could employ multi-country samples and incorporate cultural dimensions as moderating variables.

Third, while our study examined digital employee ethical anxiety as a key moderator, other organizational factors merit investigation. Future scholars could explore how leadership styles (e.g., digital leadership, sustainable leadership) and organizational culture (e.g., innovation culture, sustainability orientation) influence the effectiveness of digital employee adoption in retail settings. Additionally, investigating the moderating roles of market dynamism, competitive intensity, and regulatory pressure could provide valuable insights into the boundary conditions of our findings.

Fourth, our focus on organizational-level outcomes could be expanded to include multiple stakeholder perspectives. Future studies should examine how customers, employees, suppliers, and other stakeholders perceive and interact with digital employees in retail settings. Research could investigate how these various stakeholder interactions collectively influence sustainable innovation outcomes and long-term competitive advantage.

Finally, technological advancement in AI and digital employees continues rapidly. Future research should explore how emerging technologies (e.g., generative AI, metaverse) reshape retail operations and sustainable innovation practices. Longitudinal studies tracking the co-evolution of digital capabilities and sustainable innovation could provide valuable insights for theory and practice.

8. Conclusion

This study significantly advances the discourse on digital transformation and sustainable innovation in retail and consumer services by introducing the novel Adoption-Innovations-Future Performance (AIF) model. Extending the traditional Antecedents-Behavior-Consequences (ABC) framework, the AIF model uniquely integrates digital employee adoption, ambidextrous green innovation—both exploitative and exploratory—and future sustainable innovation performance within the context of online retailing. Our findings validate this model, demonstrating that digital employees substantially enhance retailers' capabilities for sustainable innovation in customer-facing operations. Both exploitative and exploratory green innovations emerge as crucial for achieving long-term sustainability goals, with exploratory innovation exerting a stronger influence on future performance. Notably, the study uncovers the paradoxical role of digital employee ethical anxiety: while it impedes green innovation efforts, it simultaneously strengthens the positive relationship between digital employee adoption and sustainable innovation performance in retail settings. This nuanced understanding challenges existing assumptions in the literature on digital transformation, sustainable innovation, and AI ethics, adding depth to the academic discourse. By filling critical gaps and providing both empirical evidence and theoretical advancements—such as conceptualizing digital employees as quasi-team members and offering a configurational perspective through fsQCA—our research offers actionable insights for retailers and policymakers. These insights include strategies for integrating digital employees while balancing technological advancements with ethical considerations, ultimately demonstrating the real-world significance of our study. In summary, the AIF model and our comprehensive analysis not only extend existing theories but also provide multiple pathways to achieving high sustainable innovation

performance, thereby contributing to both scholarly understanding and practical strategies for leveraging digital technologies for sustainability in retail operations.

CRediT authorship contribution statement

Shaofeng Wang: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Conceptualization. **Hao Zhang:** Writing – review & editing, Writing – original draft.

Data availability

The datasets used or analysed during the current study are available from the author on reasonable request.

Ethics approval

Not applicable.

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Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled “*Artificial Intelligence Digital Employees and Sustainable Innovation in Online Retail: The Mediating Role of Ambidextrous Green Innovation and the Moderating Role of Ethical Anxiety*”.

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Data availability

Data will be made available on request.

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